STAR Grafana Dashboard JSON

Source Listing

Auto-generated syntax-highlighted listing

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STAR Development Team
Locked Inc.

Generated by `scripts/codedump-pdf.sh`.

Contents

1 (root)

grafana-star-dashboard.json


```json
{
  "apiVersion": "dashboard.grafana.app/v2",
  "kind": "Dashboard",
  "metadata": {
    "name": "star-robot-telemetry",
    "namespace": "default",
    "uid": "3263aaa7-51c4-4d1a-8786-c86f962177bc",
    "resourceVersion": "1776950319890995",
    "generation": 25,
    "creationTimestamp": "2026-04-23T07:12:11Z",
    "labels": {
      "grafana.app/deprecatedInternalID": "2577689642606592"
    },
    "annotations": {
      "grafana.app/createdBy": "user:dfi7zlylxemtce",
      "grafana.app/message": "STOP%: debounced live-push on input change (no need to click Apply)",
      "grafana.app/updatedBy": "user:dfi7zlylxemtce",
      "grafana.app/updatedTimestamp": "2026-04-23T13:18:39Z",
      "grafana.app/folder": ""
    }
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  "spec": {
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        "kind": "AnnotationQuery",
        "spec": {
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            "group": "grafana",
            "version": "v0",
            "datasource": {
              "name": "-- Grafana --"
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            "spec": {}
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          "enable": true,
          "hide": true,
          "iconColor": "rgba(0, 211, 255, 1)",
          "name": "Annotations & Alerts",
          "builtIn": true,
          "legacyOptions": {
            "type": "dashboard"
          }
        }
      }
    ],
    "cursorSync": "Crosshair",
    "editable": true,
    "elements": {
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        "spec": {
          "id": 1,
          "title": "FL velocity",
          "description": "",
          "links": [],
          "data": {
            "kind": "QueryGroup",
            "spec": {
              "queries": [
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                  "kind": "PanelQuery",
                  "spec": {
                    "query": {
                      "kind": "DataQuery",
                      "group": "",
                      "version": "v0",
                      "spec": {
                        "expr": "star_motor_velocity_mps{wheel=\"fl\"}",
                        "legendFormat": ""
                      }
                    },
                    "refId": "A",
                    "hidden": false
                  }
```

```
                  }
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                "transformations": [],
                "queryOptions": {}
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            "vizConfig": {
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              "group": "gauge",
              "version": "",
              "spec": {
                "options": {
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                  "showThresholdLabels": false,
                  "showThresholdMarkers": true
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                "fieldConfig": {
                  "defaults": {
                    "unit": "velocityms",
                    "min": -0.3,
                    "max": 0.3,
                    "thresholds": {
                      "mode": "absolute",
                      "steps": [
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                          "value": null,
                          "color": "blue"
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                        {
                          "value": -0.25,
                          "color": "green"
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                        {
                          "value": 0.2,
                          "color": "yellow"
                        },
                        {
                          "value": 0.25,
                          "color": "red"
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                  "overrides": []
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              }
            }
          }
        },
        "panel-10": {
          "kind": "Panel",
          "spec": {
            "id": 10,
            "title": "Wheel duty over time",
            "description": "",
            "links": [],
            "data": {
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              "spec": {
                "queries": [
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                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_motor_duty_percent",
                          "legendFormat": "{{wheel}}"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
```

```
              }
            },
            "vizConfig": {
              "kind": "VizConfig",
              "group": "timeseries",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
                  "defaults": {
                    "unit": "percent"
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-11": {
          "kind": "Panel",
          "spec": {
            "id": 11,
            "title": "Motor faults (any != 0 is bad)",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
                "queries": [
                  {
                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_motor_fault_byte",
                          "legendFormat": "{{wheel}}"
                        }
                      },
                      "refId": "A",
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                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
              }
            },
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              "kind": "VizConfig",
              "group": "stat",
              "version": "",
              "spec": {
                "options": {
                  "colorMode": "background",
                  "graphMode": "none",
                  "textMode": "value_and_name"
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                "fieldConfig": {
                  "defaults": {
                    "thresholds": {
                      "mode": "absolute",
                      "steps": [
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                          "value": null,
                          "color": "green"
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                        {
                          "value": 1,
                          "color": "red"
                        }
                      ]
                    }
                  },
                  "overrides": []
                }
              }
```

```
            }
          }
        },
        "panel-12": {
          "kind": "Panel",
          "spec": {
            "id": 12,
            "title": "Roll",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
                "queries": [
                  {
                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_imu_orientation_deg{axis=\"roll\"}"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
              }
            },
            "vizConfig": {
              "kind": "VizConfig",
              "group": "gauge",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
                  "defaults": {
                    "unit": "degree",
                    "min": -180,
                    "max": 180
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-13": {
          "kind": "Panel",
          "spec": {
            "id": 13,
            "title": "Pitch",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
                "queries": [
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                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_imu_orientation_deg{axis=\"pitch\"}"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
```

```
                "transformations": [],
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            },
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              "group": "gauge",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
                  "defaults": {
                    "unit": "degree",
                    "min": -180,
                    "max": 180
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-14": {
          "kind": "Panel",
          "spec": {
            "id": 14,
            "title": "Heading",
            "description": "",
            "links": [],
            "data": {
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              "spec": {
                "queries": [
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                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_imu_orientation_deg{axis=\"heading\"}"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
              }
            },
            "vizConfig": {
              "kind": "VizConfig",
              "group": "gauge",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
                  "defaults": {
                    "unit": "degree",
                    "min": 0,
                    "max": 360
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-15": {
          "kind": "Panel",
          "spec": {
            "id": 15,
            "title": "Gyro (deg/s)",
            "description": "",
            "links": [],
            "data": {
```

```
            "kind": "QueryGroup",
            "spec": {
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                  "spec": {
                    "query": {
                      "kind": "DataQuery",
                      "group": "",
                      "version": "v0",
                      "spec": {
                        "expr": "star_imu_gyro_dps",
                        "legendFormat": "{{axis}}"
                      }
                    },
                    "refId": "A",
                    "hidden": false
                  }
                }
              ],
              "transformations": [],
              "queryOptions": {}
            }
          },
          "vizConfig": {
            "kind": "VizConfig",
            "group": "timeseries",
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            "spec": {
              "options": {},
              "fieldConfig": {
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                  "unit": "rotdegs"
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                "overrides": []
              }
            }
          }
        }
      },
      "panel-16": {
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        "spec": {
          "id": 16,
          "title": "Linear acceleration (m/s^2)",
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          "links": [],
          "data": {
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            "spec": {
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                  "spec": {
                    "query": {
                      "kind": "DataQuery",
                      "group": "",
                      "version": "v0",
                      "spec": {
                        "expr": "star_imu_accel_mps2",
                        "legendFormat": "{{axis}}"
                      }
                    },
                    "refId": "A",
                    "hidden": false
                  }
                }
              ],
              "transformations": [],
              "queryOptions": {}
            }
          },
          "vizConfig": {
            "kind": "VizConfig",
            "group": "timeseries",
            "version": "",
            "spec": {
              "options": {},
              "fieldConfig": {
```

```
                "defaults": {
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                "overrides": []
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            }
          }
        }
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          "title": "Linear velocity",
          "description": "",
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            "spec": {
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                  "spec": {
                    "query": {
                      "kind": "DataQuery",
                      "group": "",
                      "version": "v0",
                      "spec": {
                        "expr": "star_odom_linear_mps"
                      }
                    },
                    "refId": "A",
                    "hidden": false
                  }
                }
              ],
              "transformations": [],
              "queryOptions": {}
            }
          },
          "vizConfig": {
            "kind": "VizConfig",
            "group": "gauge",
            "version": "",
            "spec": {
              "options": {},
              "fieldConfig": {
                "defaults": {
                  "unit": "velocityms",
                  "min": -0.3,
                  "max": 0.3
                },
                "overrides": []
              }
            }
          }
        }
      },
      "panel-18": {
        "kind": "Panel",
        "spec": {
          "id": 18,
          "title": "Angular velocity",
          "description": "",
          "links": [],
          "data": {
            "kind": "QueryGroup",
            "spec": {
              "queries": [
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                  "spec": {
                    "query": {
                      "kind": "DataQuery",
                      "group": "",
                      "version": "v0",
                      "spec": {
                        "expr": "star_odom_angular_radps"
                      }
```

```
              },
              "refId": "A",
              "hidden": false
            }
          }
        ],
        "transformations": [],
        "queryOptions": {}
      }
    },
    "vizConfig": {
      "kind": "VizConfig",
      "group": "gauge",
      "version": "",
      "spec": {
        "options": {},
        "fieldConfig": {
          "defaults": {
            "unit": "rads",
            "min": -2,
            "max": 2
          },
          "overrides": []
        }
      }
    }
  }
},
"panel-19": {
  "kind": "Panel",
  "spec": {
    "id": 19,
    "title": "Odom pose (x, y, theta)",
    "description": "",
    "links": [],
    "data": {
      "kind": "QueryGroup",
      "spec": {
        "queries": [
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            "kind": "PanelQuery",
            "spec": {
              "query": {
                "kind": "DataQuery",
                "group": "",
                "version": "v0",
                "spec": {
                  "expr": "star_odom_pose_x_m",
                  "legendFormat": "x (m)"
                }
              },
              "refId": "A",
              "hidden": false
            }
          },
          {
            "kind": "PanelQuery",
            "spec": {
              "query": {
                "kind": "DataQuery",
                "group": "",
                "version": "v0",
                "spec": {
                  "expr": "star_odom_pose_y_m",
                  "legendFormat": "y (m)"
                }
              },
              "refId": "B",
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            }
          },
          {
            "kind": "PanelQuery",
            "spec": {
              "query": {
                "kind": "DataQuery",
                "group": "",
                "version": "v0",
                "spec": {
```

```
                          "expr": "star_odom_pose_theta_rad",
                          "legendFormat": "theta (rad)"
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                      "refId": "C",
                      "hidden": false
                    }
                  }
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                "queryOptions": {}
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            },
            "vizConfig": {
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              "group": "stat",
              "version": "",
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                  "graphMode": "area",
                  "textMode": "value_and_name"
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                "fieldConfig": {
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                  "overrides": []
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              }
            }
          }
        },
        "panel-2": {
          "kind": "Panel",
          "spec": {
            "id": 2,
            "title": "FR velocity",
            "description": "",
            "links": [],
            "data": {
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              "spec": {
                "queries": [
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                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_motor_velocity_mps{wheel=\"fr\"}",
                          "legendFormat": ""
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                      },
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                "transformations": [],
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            },
            "vizConfig": {
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              "group": "gauge",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
                  "defaults": {
                    "unit": "velocityms",
                    "min": -0.3,
                    "max": 0.3,
                    "thresholds": {
                      "mode": "absolute",
                      "steps": [
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                          "value": null,
```

```
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                        {
                          "value": -0.25,
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                        {
                          "value": 0.2,
                          "color": "yellow"
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                        {
                          "value": 0.25,
                          "color": "red"
                        }
                      ]
                    }
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-20": {
          "kind": "Panel",
          "spec": {
            "id": 20,
            "title": "Line rate by kind (msg/s)",
            "description": "",
            "links": [],
            "data": {
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              "spec": {
                "queries": [
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                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "rate(star_bridge_rx_lines_total[1m])",
                          "legendFormat": "{{kind}}"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
              }
            },
            "vizConfig": {
              "kind": "VizConfig",
              "group": "timeseries",
              "version": "",
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                "fieldConfig": {
                  "defaults": {},
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-21": {
          "kind": "Panel",
          "spec": {
            "id": 21,
            "title": "Parse errors (/s)",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
```

```
                "queries": [
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                    "kind": "PanelQuery",
                    "spec": {
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                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "rate(star_bridge_parse_errors_total[1m])",
                          "legendFormat": "{{kind}}"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
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                "transformations": [],
                "queryOptions": {}
              }
            },
            "vizConfig": {
              "kind": "VizConfig",
              "group": "timeseries",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
                  "defaults": {
                    "color": {
                      "mode": "fixed",
                      "fixedColor": "red"
                    }
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-22": {
          "kind": "Panel",
          "spec": {
            "id": 22,
            "title": "Serial reopens (cumulative)",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
                "queries": [
                  {
                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_bridge_serial_reopens_total"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
              }
            },
            "vizConfig": {
              "kind": "VizConfig",
              "group": "stat",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
```

```
                  "defaults": {},
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-23": {
          "kind": "Panel",
          "spec": {
            "id": 23,
            "title": "RX bytes total",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
                "queries": [
                  {
                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_bridge_rx_bytes_total"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
              }
            },
            "vizConfig": {
              "kind": "VizConfig",
              "group": "stat",
              "version": "",
              "spec": {
                "options": {},
                "fieldConfig": {
                  "defaults": {
                    "unit": "bytes"
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-24": {
          "kind": "Panel",
          "spec": {
            "id": 24,
            "title": "Closest obstacle",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
                "queries": [
                  {
                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_scan_closest_range_m"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
```

```
              }
            ],
            "transformations": [],
            "queryOptions": {}
          }
        },
        "vizConfig": {
          "kind": "VizConfig",
          "group": "gauge",
          "version": "",
          "spec": {
            "options": {},
            "fieldConfig": {
              "defaults": {
                "unit": "lengthm",
                "min": 0,
                "max": 3,
                "thresholds": {
                  "mode": "absolute",
                  "steps": [
                    {
                      "value": null,
                      "color": "red"
                    },
                    {
                      "value": 0.5,
                      "color": "yellow"
                    },
                    {
                      "value": 1,
                      "color": "green"
                    }
                  ]
                }
              },
              "overrides": []
            }
          }
        }
      }
    },
    "panel-25": {
      "kind": "Panel",
      "spec": {
        "id": 25,
        "title": "Closest bearing (0deg=forward)",
        "description": "",
        "links": [],
        "data": {
          "kind": "QueryGroup",
          "spec": {
            "queries": [
              {
                "kind": "PanelQuery",
                "spec": {
                  "query": {
                    "kind": "DataQuery",
                    "group": "",
                    "version": "v0",
                    "spec": {
                      "expr": "star_scan_closest_bearing_deg"
                    }
                  },
                  "refId": "A",
                  "hidden": false
                }
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            ],
            "transformations": [],
            "queryOptions": {}
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        },
        "vizConfig": {
          "kind": "VizConfig",
          "group": "gauge",
          "version": "",
          "spec": {
            "options": {},
            "fieldConfig": {
```

```
                  "defaults": {
                    "unit": "degree",
                    "min": -180,
                    "max": 180
                  },
                  "overrides": []
                }
              }
            }
          }
        },
        "panel-26": {
          "kind": "Panel",
          "spec": {
            "id": 26,
            "title": "Obstacles < 1 m",
            "description": "",
            "links": [],
            "data": {
              "kind": "QueryGroup",
              "spec": {
                "queries": [
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                    "kind": "PanelQuery",
                    "spec": {
                      "query": {
                        "kind": "DataQuery",
                        "group": "",
                        "version": "v0",
                        "spec": {
                          "expr": "star_scan_obstacles_under_1m"
                        }
                      },
                      "refId": "A",
                      "hidden": false
                    }
                  }
                ],
                "transformations": [],
                "queryOptions": {}
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            },
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          }
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                          "legendFormat": "{{stat}}"
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            "links": [],
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```

```
            "content":
"<style>\n.star-cockpit{display:flex;flex-direction:column;height:100%;gap:8px;background:#111;padding:8px;box-sizing:borde
b{font:600 12px system-ui;color:#888;margin-right:4px;min-width:72px}\n.star-btn{padding:6px
12px;border-radius:4px;border:1px solid #444;background:#222;color:#ccc;text-decoration:none;font:600 13px
system-ui;cursor:pointer;white-space:nowrap;user-select:none}\n.star-btn:hover{background:#333}\n.star-btn.danger{backgroun
10px;border-radius:20px;border:1px solid
#444;background:#222;cursor:pointer;user-select:none;min-width:130px;justify-content:space-between}\n.star-switch:hover{bac
.knob{width:38px;height:20px;border-radius:12px;background:#444;position:relative;transition:background
.15s}\n.star-switch
.knob::after{content:\"\";position:absolute;top:2px;left:2px;width:16px;height:16px;border-radius:50%;background:#ddd;trans
.15s}\n.star-switch.on .knob{background:#0a6}\n.star-switch.on
.knob::after{left:20px}\n.star-switch.estop.on .knob{background:#c22}\n.star-switch .label{font:600 13px
system-ui}\n.star-switch.on.estop .label{color:#f88}\n.star-switch.on:not(.estop)
.label{color:#8f8}\n.star-num{display:inline-flex;align-items:center;gap:6px}\n.star-num
input[type=number]{width:80px;padding:4px 6px;border-radius:4px;border:1px solid
#444;background:#222;color:#eee;font:13px system-ui}\n.star-num span{font:600 11px
system-ui;color:#888}\n.star-main{flex:1;display:flex;gap:8px;min-height:0}\n.star-map-img{flex:1;display:flex;justify-cont
img{max-width:100%;max-height:100%;image-rendering:pixelated}\n.star-map-img
img[src=\"\"]{display:none}\n.star-teleop{display:grid;grid-template-columns:60px 60px
60px;grid-template-rows:60px 60px 60px;gap:4px;align-self:center}\n.star-tele-btn{background:#222;border:1px
solid #444;border-radius:6px;color:#ccc;font:700 20px
system-ui;cursor:pointer;user-select:none;display:flex;align-items:center;justify-content:center}\n.star-tele-btn:hover{bac
(pointer: coarse), (max-width: 820px){ .star-tele-btn.corner{display:none} }\n#star-status{font:12px
monospace;color:#888;margin-left:auto}\n.star-hint{font:11px
system-ui;color:#666;margin-left:8px}\n</style>\n<div class=\"star-cockpit\" tabindex=\"0\"
id=\"star-cockpit\">\n  <div class=\"star-row\">\n    <b>CONTROL</b>\n    <label class=\"star-switch estop\"
id=\"star-sw-estop\"><span class=\"label\">E-STOP</span><span class=\"knob\"></span></label>\n    <label
class=\"star-switch\" id=\"star-sw-auto\"><span class=\"label\">MANUAL</span><span
class=\"knob\"></span></label>\n    <span class=\"star-num\">\n      <span>SPEED</span>\n      <input
type=\"number\" id=\"star-speed\" min=\"0\" max=\"100\" step=\"5\" value=\"50\">\n      <span>%</span>\n
<button class=\"star-btn\" id=\"star-speed-apply\">Apply</button>\n    </span>\n    <span
class=\"star-num\">\n      <span>TURN</span>\n      <input type=\"number\" id=\"star-turn\" min=\"0.5\"
max=\"8\" step=\"0.5\" value=\"4\">\n      <span>rad/s</span>\n      <button class=\"star-btn\"
id=\"star-turn-apply\">Apply</button>\n    </span>\n    <span class=\"star-hint\">click panel then WASD /
arrows to drive</span>\n    <span id=\"star-status\">loading...</span>\n  </div>\n  <div
class=\"star-row\">\n    <b>MAP</b>\n    <a class=\"star-btn\"
href=\"https://bsikar:y0hj8pQBG1zbpyrDTAG9UK@robot.flavorblast.sbs/map.pgm\" download=\"map.pgm\">Download
.pgm</a>\n    <a class=\"star-btn\"
href=\"https://bsikar:y0hj8pQBG1zbpyrDTAG9UK@robot.flavorblast.sbs/map.yaml\" download=\"map.yaml\">Download
.yaml</a>\n    <button class=\"star-btn\" id=\"star-map-record\" title=\"Start/stop SLAM
mapping\">loading...</button>\n    <button class=\"star-btn danger\" id=\"star-map-clear\" title=\"Wipe the
map (preserves current mapping state)\">CLEAR MAP</button>\n    <button class=\"star-btn\"
id=\"star-explore\" title=\"Auto-explore the area until no frontiers remain (requires mapping
active)\">loading...</button>\n    <span class=\"star-num\" title=\"How patient to be before declaring
exploration done (higher = wait longer for growth)\">\n      <span>STOP%</span>\n      <input
type=\"number\" id=\"star-explore-th\" min=\"0\" max=\"100\" step=\"5\" value=\"50\">\n
<span>%</span>\n      <button class=\"star-btn\" id=\"star-explore-th-apply\">Apply</button>\n    </span>\n
</div>\n  <div class=\"star-row\" style=\"font:11px system-ui;color:#9a9;padding:2px 0;border-left:2px solid
#363;padding-left:10px;gap:18px\">\n    <b style=\"color:#666\">HINTS</b>\n    <span><b
style=\"color:#bc8\">MANUAL (WASD/arrows):</b> SPEED 50% first / 100% comfy * TURN 3-4 rad/s</span>\n
<span><b style=\"color:#8bc\">AUTO goals (click in 3D):</b> SPEED 60% * TURN 1-2 rad/s</span>\n    <span><b
style=\"color:#c8b\">EXPLORE (auto-map):</b> start MAPPING first, then EXPLORE. STOP% = patience (higher
waits longer before calling it done)</span>\n  </div>\n<div class=\"star-main\">\n    <div
class=\"star-map-img\"><img id=\"star-map-img\" src=\"\" alt=\"\"></div>\n    <div class=\"star-teleop\"
id=\"star-teleop\">\n      <div class=\"star-tele-btn ne corner\" data-dir=\"up-left\">NW</div>\n      <div
class=\"star-tele-btn up\" data-dir=\"up\">N</div>\n      <div class=\"star-tele-btn nw corner\"
data-dir=\"up-right\">NE</div>\n      <div class=\"star-tele-btn left\" data-dir=\"left\">W</div>\n
<div class=\"star-tele-btn right\" data-dir=\"right\">E</div>\n      <div class=\"star-tele-btn sw corner\"
data-dir=\"down-left\">SW</div>\n      <div class=\"star-tele-btn down\" data-dir=\"down\">S</div>\n
<div class=\"star-tele-btn se corner\" data-dir=\"down-right\">SE</div>\n    </div>\n
</div>\n</div>\n<script>\n(function(){\n  const AUTH = \"Basic YnNpa2FyOnkwaGo4cFFCRzF6YnB5ckRUQUc5VUs=\";\n
const _fetch = window.fetch.bind(window);\n  function fetch(url, opts){ opts = opts||{}; opts.headers =
Object.assign({}, opts.headers||{}, {Authorization: AUTH}); return _fetch(url, opts); }\n  const API =
\"https://robot.flavorblast.sbs/api\";\n  const PUBLIC = \"https://robot.flavorblast.sbs\";\n  const LINEAR
= 0.8, HALF_BASE = 0.178, MAX_WHEEL = 0.9, RATE_MS = 100;\n\n  // ANGULAR: start from localStorage or
default 4.0\n  const LS_ANG = \"starCockpit.angular\";\n  let ANGULAR =
parseFloat(localStorage.getItem(LS_ANG));\n  if (!(ANGULAR > 0)) ANGULAR = 4.0;\n\n  // DOM\n  const img =
document.getElementById(\"star-map-img\");\n  const status = document.getElementById(\"star-status\");\n
const swEstop = document.getElementById(\"star-sw-estop\");\n  const swAuto =
document.getElementById(\"star-sw-auto\");\n  const spInput = document.getElementById(\"star-speed\");\n
const spApply = document.getElementById(\"star-speed-apply\");\n  const trInput =
document.getElementById(\"star-turn\");\n  const trApply = document.getElementById(\"star-turn-apply\");\n
const btnRecord = document.getElementById(\"star-map-record\");\n  const btnClear =
document.getElementById(\"star-map-clear\");\n  const cockpit = document.getElementById(\"star-cockpit\");\n
const teleopGrid = document.getElementById(\"star-teleop\");\n\n  trInput.value = ANGULAR;\n\n  // --- state
render ---\n  let _slamState = null;\n  function render(state){\n    if (state.estop === true) {
swEstop.classList.add(\"on\"); swEstop.querySelector(\".label\").textContent = \"E-STOPPED\"; }\n    else {
swEstop.classList.remove(\"on\"); swEstop.querySelector(\".label\").textContent = \"E-STOP\"; }\n    if
(state.autonomy === true) { swAuto.classList.add(\"on\"); swAuto.querySelector(\".label\").textContent =
\"AUTONOMY\"; }\n    else { swAuto.classList.remove(\"on\"); swAuto.querySelector(\".label\").textContent =
\"MANUAL\"; }\n    if (typeof state.speed === \"number\") spInput.placeholder =
Math.round(state.speed*100);\n  }\n  function renderRecord(){\n    if (_slamState === \"active\") {\n
btnRecord.textContent = \"[PAUSE] STOP MAPPING\";\n      btnRecord.style.background = \"#442\";
```

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```

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                "queryOptions": {}
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                }
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              "hidden": false
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                      }
                    ]
                  }
                }
              }
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```
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                        "kind": "GridLayoutItem",
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                          "height": 4,
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                        }
                      }
                    ]
                  }
                }
              }
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                "collapse": false,
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                  "spec": {
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                          "height": 7,
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                          "height": 7,
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                            "name": "panel-16"
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                    ]
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                }
              }
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                        }
                      }
                    ]
                  }
                }
              }
```

```
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              "collapse": false,
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                        }
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                        "height": 6,
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                        "element": {
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                          "name": "panel-27"
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                      }
                    }
                  ]
                }
              }
            }
          },
          {
            "kind": "RowsLayoutRow",
            "spec": {
              "title": "Bridge health",
              "collapse": true,
              "layout": {
                "kind": "GridLayout",
                "spec": {
                  "items": [
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                      "kind": "GridLayoutItem",
                      "spec": {
```

```
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                      "height": 4,
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                        "name": "panel-22"
                      }
                    }
                  },
                  {
                    "kind": "GridLayoutItem",
                    "spec": {
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                      "y": -7,
                      "width": 12,
                      "height": 4,
                      "element": {
                        "kind": "ElementReference",
                        "name": "panel-23"
                      }
                    }
                  }
                ]
              }
            }
          }
        }
      ]
    }
  },
  "links": [],
  "liveNow": false,
  "preload": false,
  "tags": [
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    "robot"
  ],
  "timeSettings": {
    "from": "now-2m",
    "to": "now",
    "autoRefresh": "1s",
    "autoRefreshIntervals": [
      "5s",
      "10s",
      "30s",
      "1m",
      "5m",
      "15m",
      "30m",
      "1h",
      "2h",
      "1d"
```

```
      ],
      "hideTimepicker": false,
      "fiscalYearStartMonth": 0
    },
    "title": "STAR robot telemetry",
    "variables": []
  }
}
```

STAR Grafana Dashboard

Rendered Snapshot

Visual representations of every panel in the `monitoring/grafana-star-dashboard.json` dashboard.

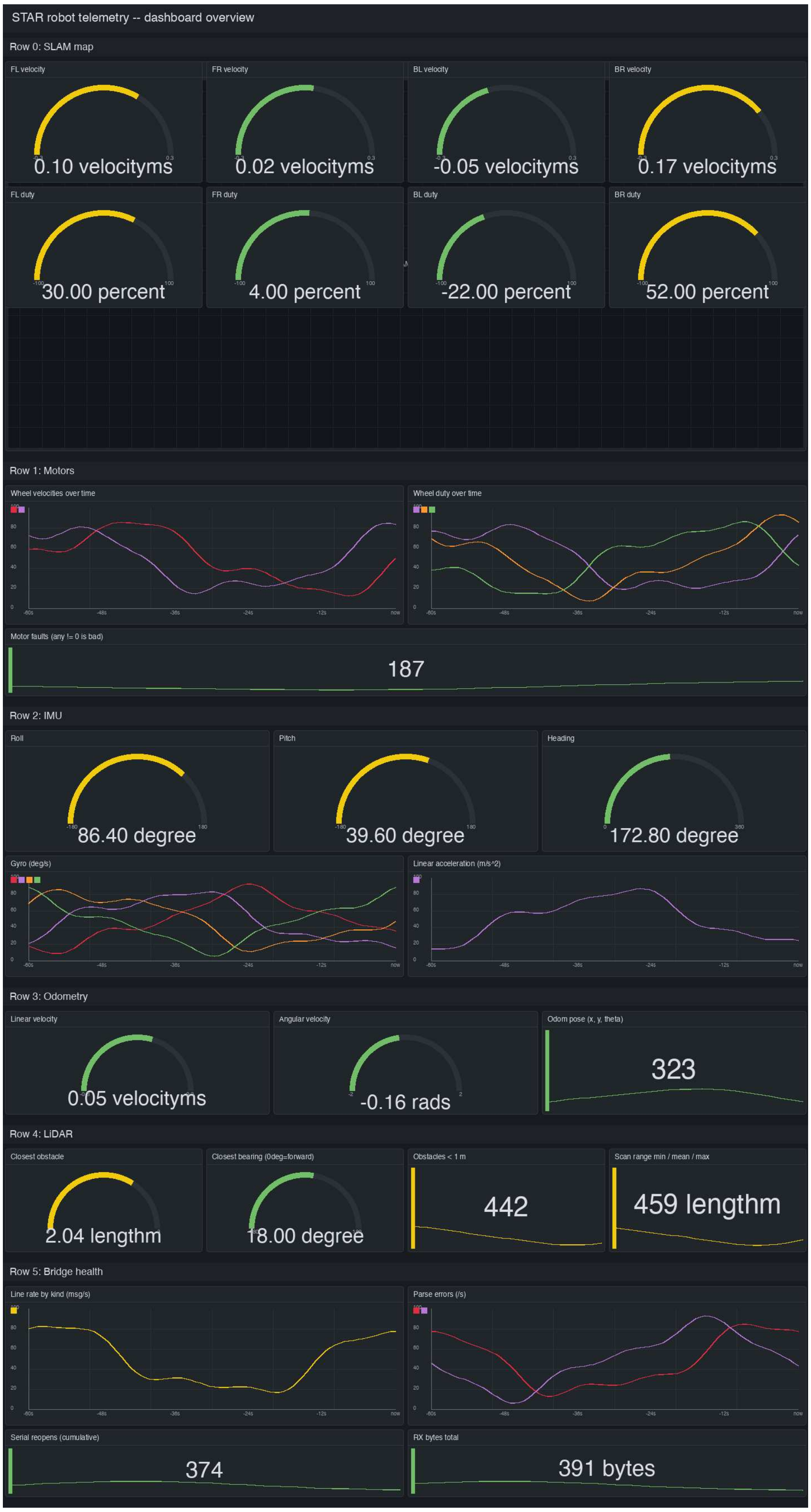
Generated by `scripts/render-grafana-json.py`
using offline PIL rendering with synthetic demo data.

To capture screenshots of the production dashboard with live Prometheus data, deploy via `monitoring/docker-compose.yml` and use the grafana-image-renderer plugin.

Dashboard Overview

STAR robot telemetry

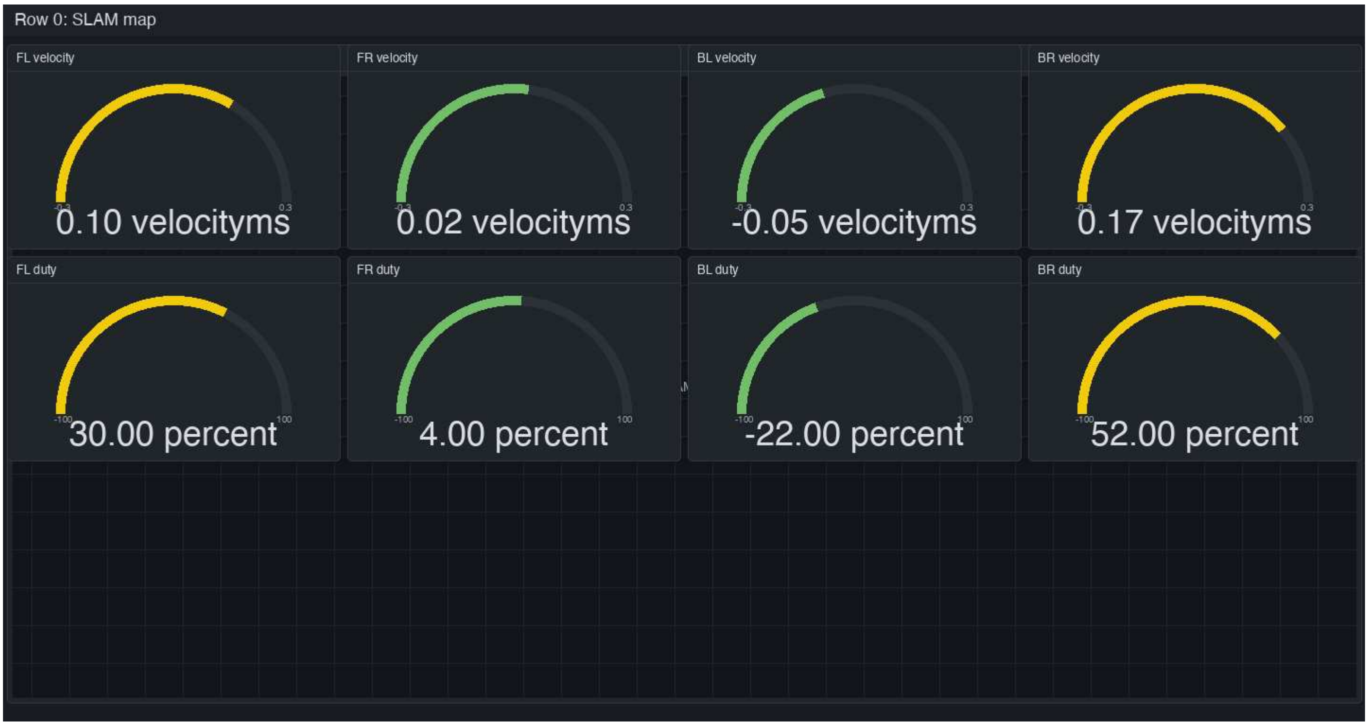
Composite of all rows



Per-Row Composites

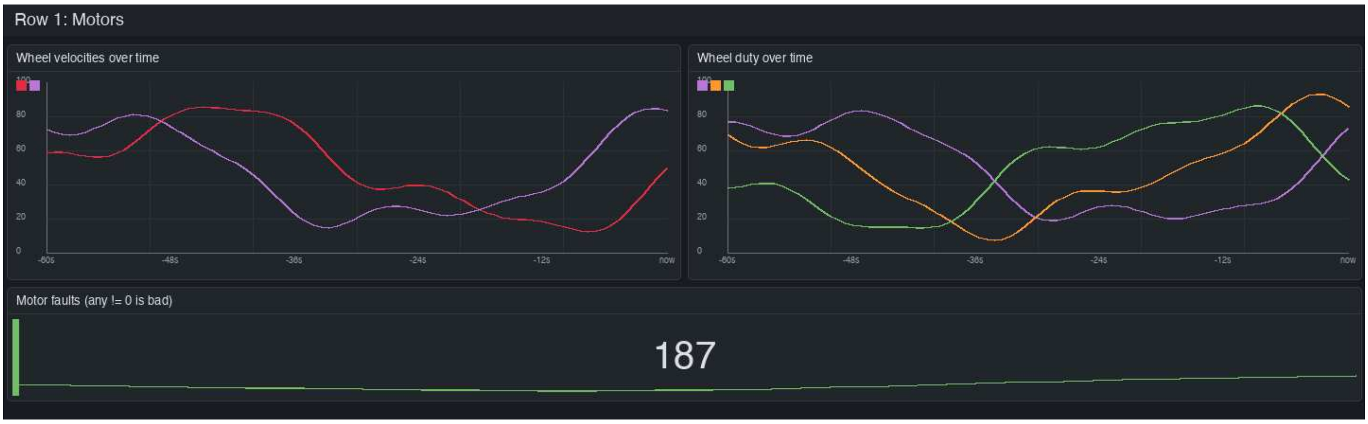
Row 0: SLAM map

All panels in this row, arranged on the Grafana 24-column grid



Row 1: Motors

All panels in this row, arranged on the Grafana 24-column grid



Row 2: IMU

All panels in this row, arranged on the Grafana 24-column grid

Row 3: Odometry

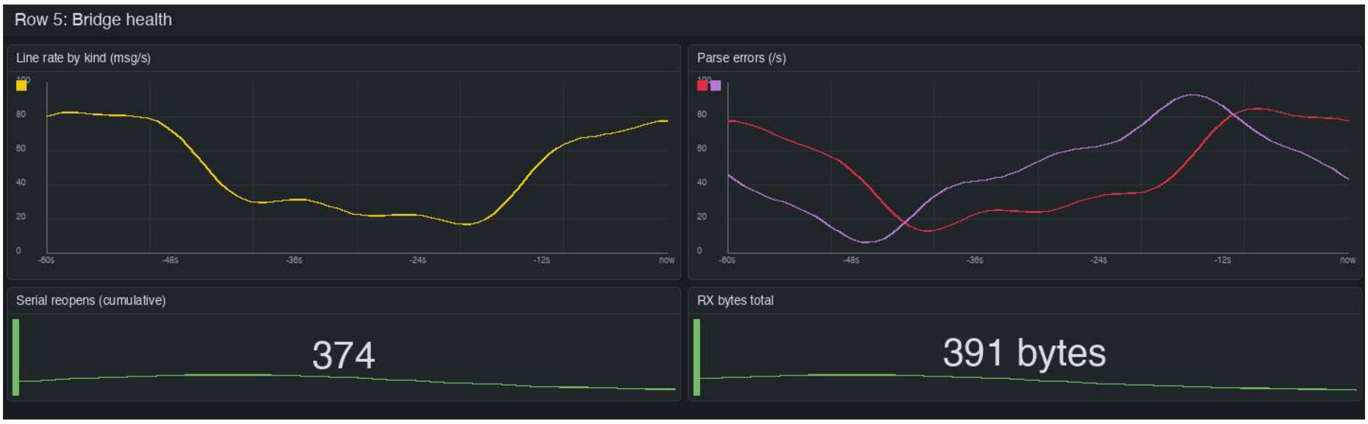
All panels in this row, arranged on the Grafana 24-column grid

Row 4: LiDAR

All panels in this row, arranged on the Grafana 24-column grid

Row 5: Bridge health

All panels in this row, arranged on the Grafana 24-column grid

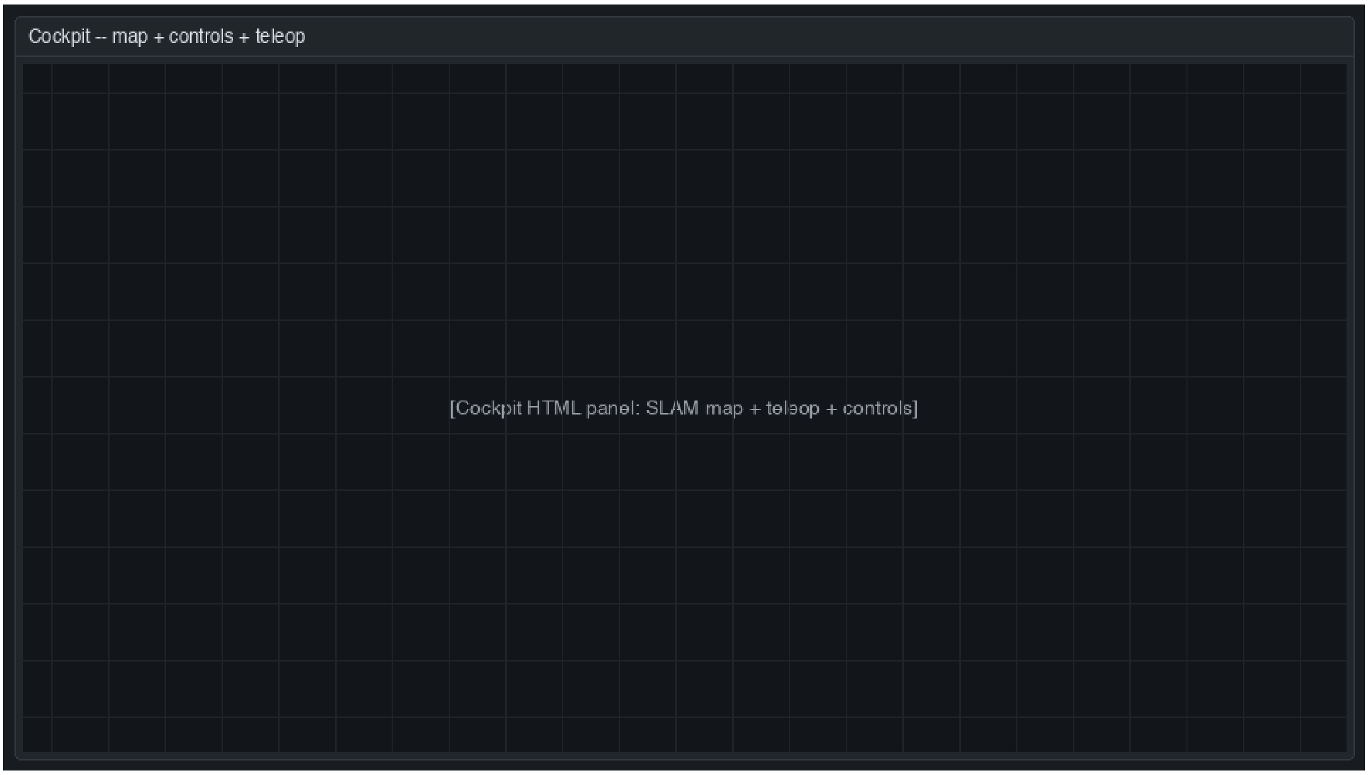


Individual Panels

Row 0: SLAM map

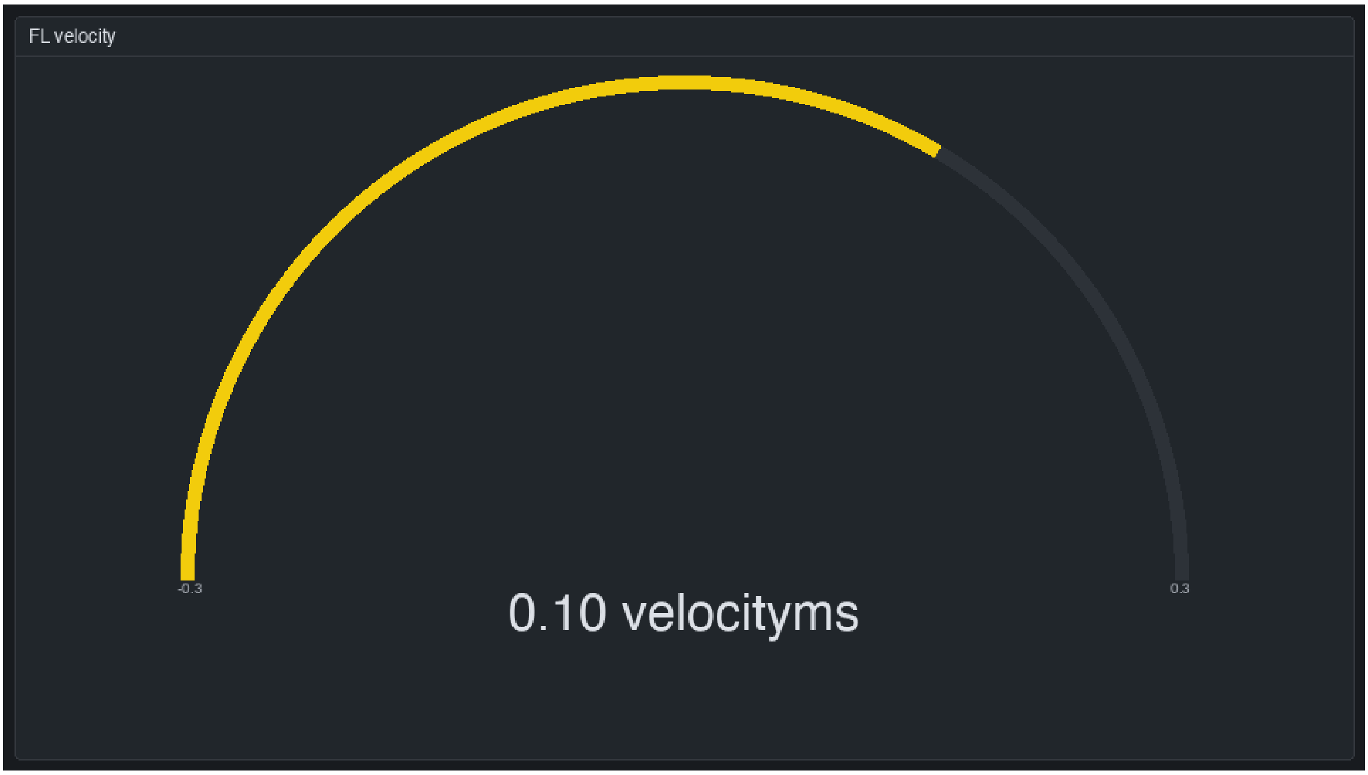
Cockpit – map + controls + teleop

Row 0: SLAM map



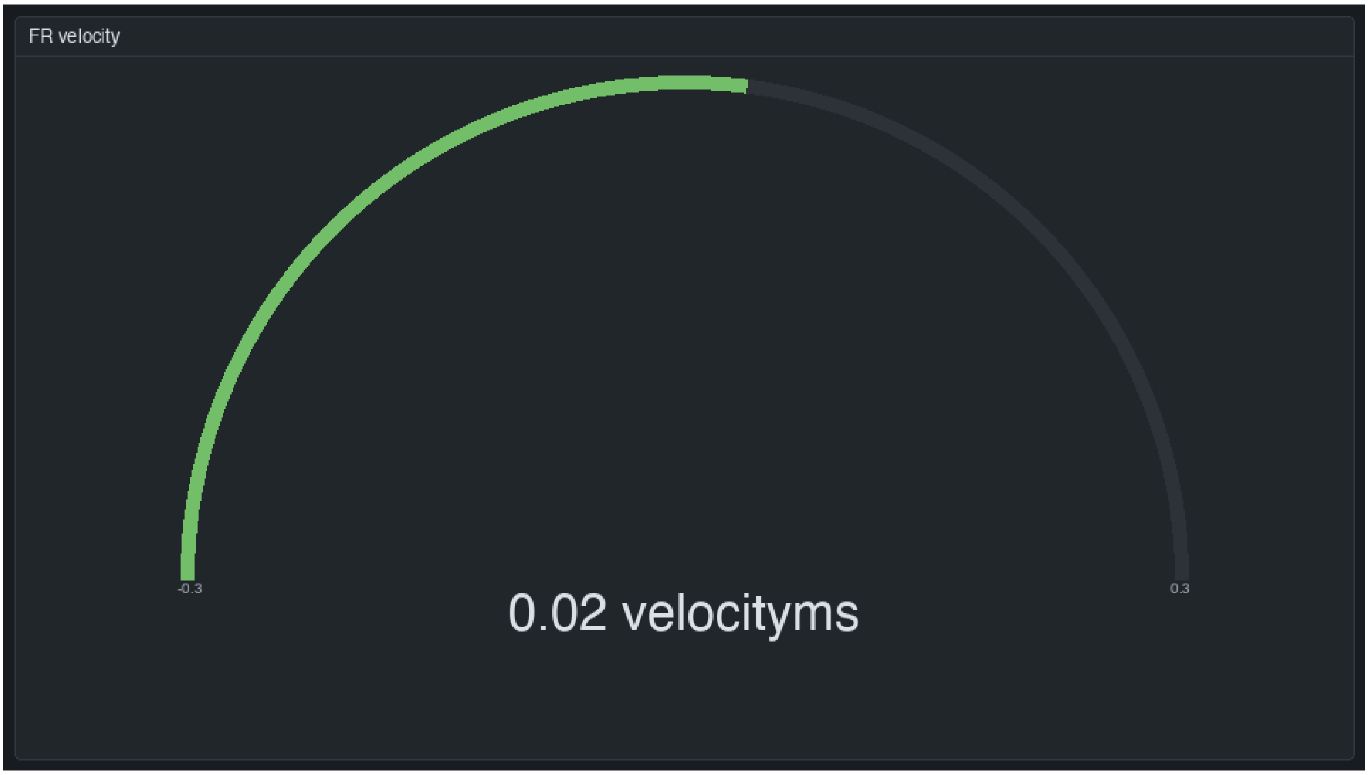
FL velocity

Row 0: SLAM map



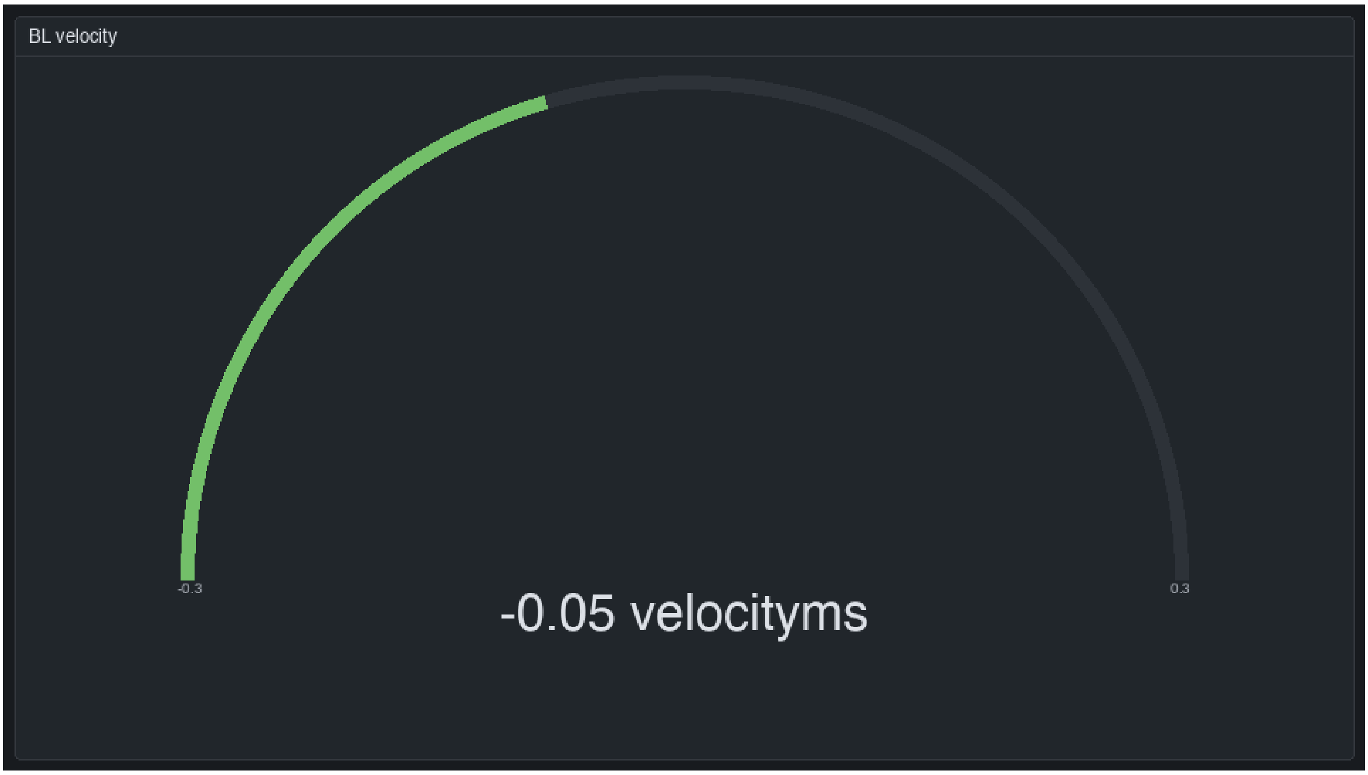
FR velocity

Row 0: SLAM map



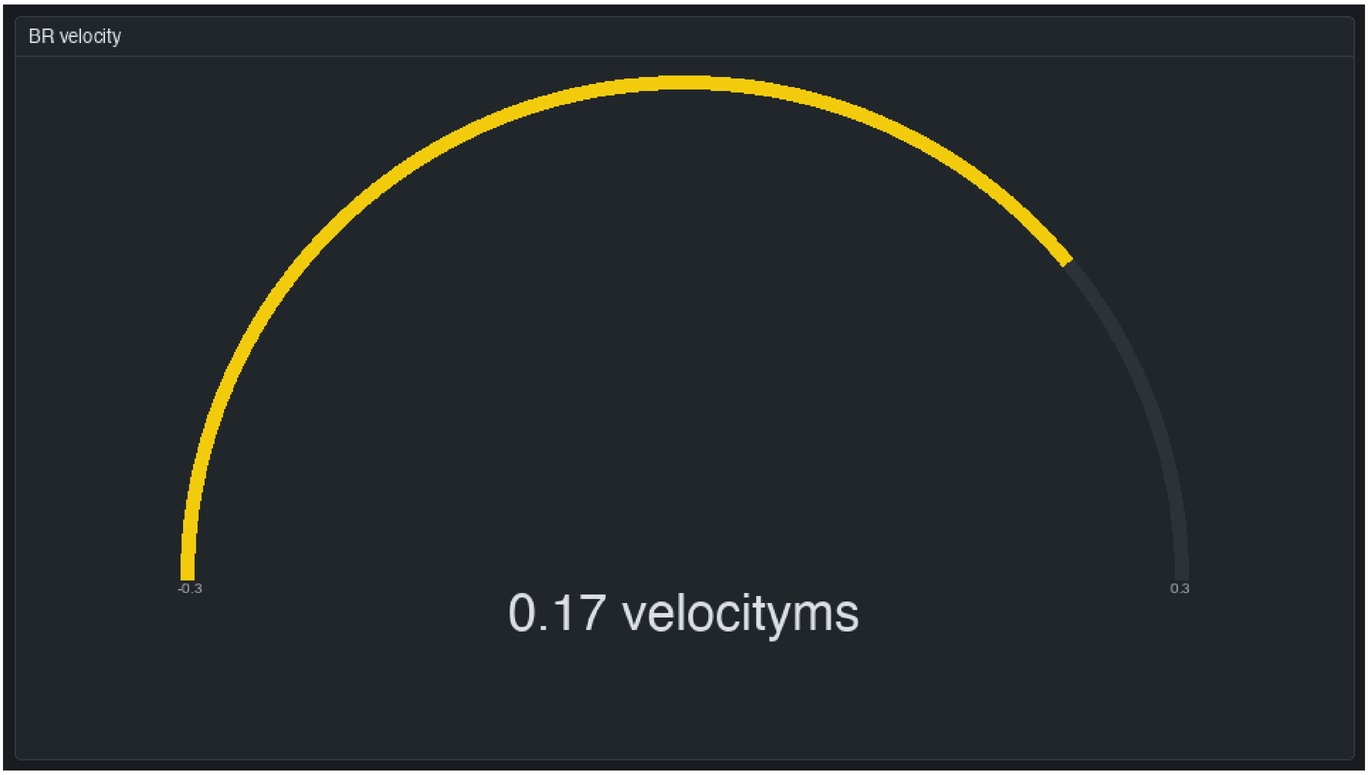
BL velocity

Row 0: SLAM map



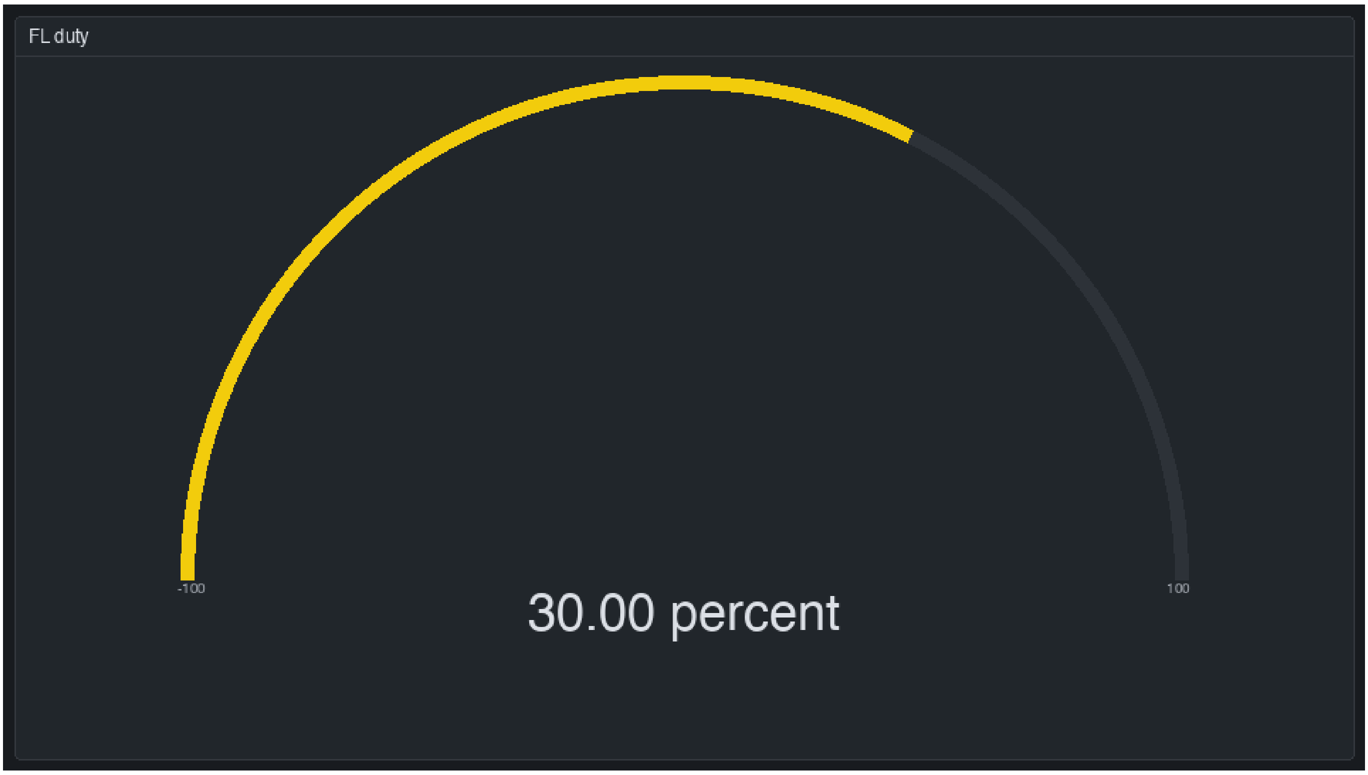
BR velocity

Row 0: SLAM map



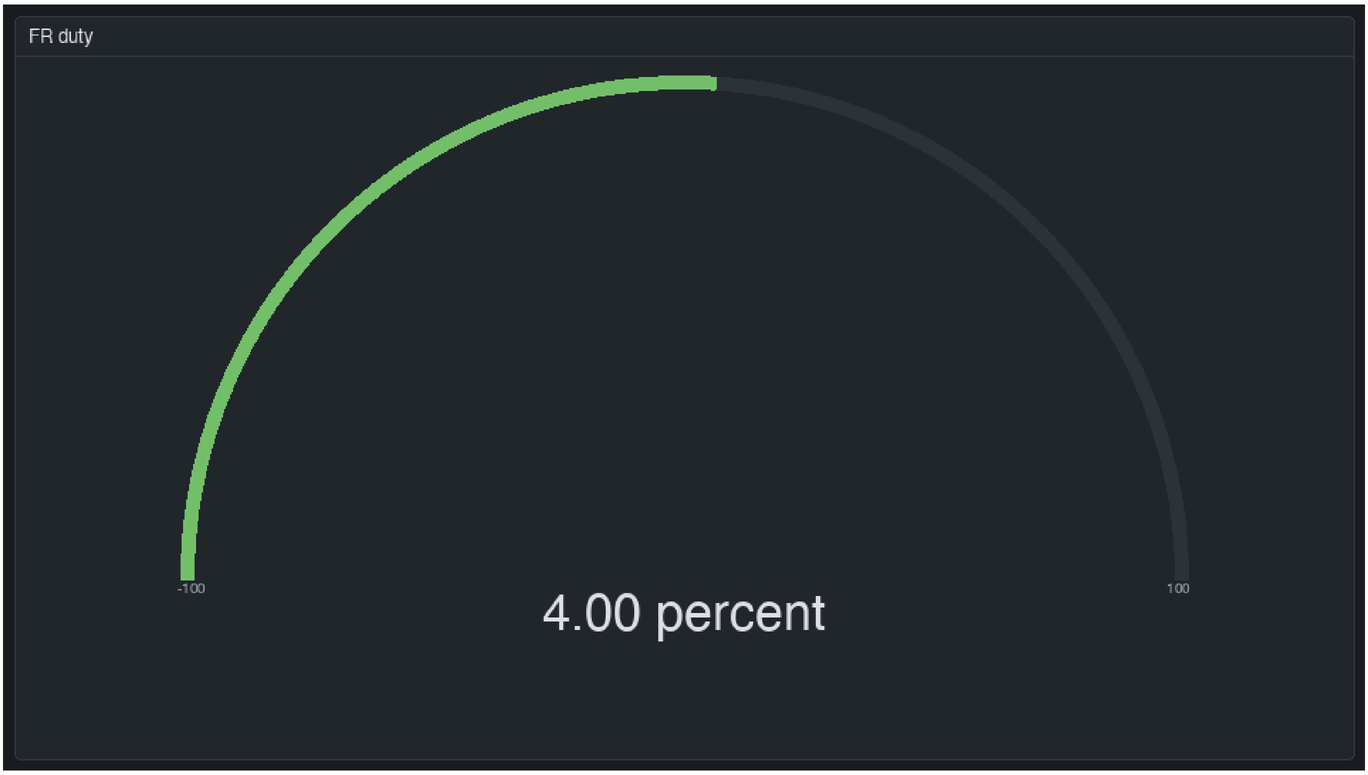
FL duty

Row 0: SLAM map



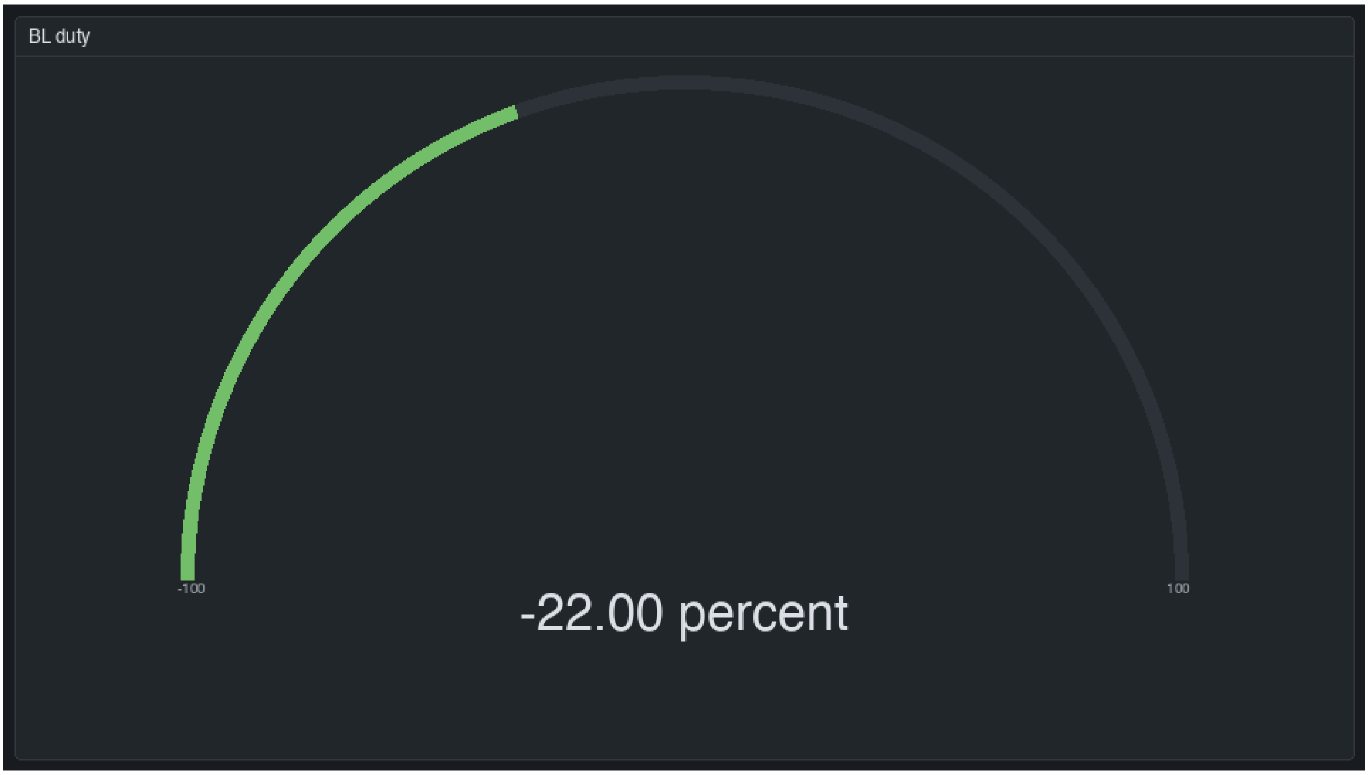
FR duty

Row 0: SLAM map



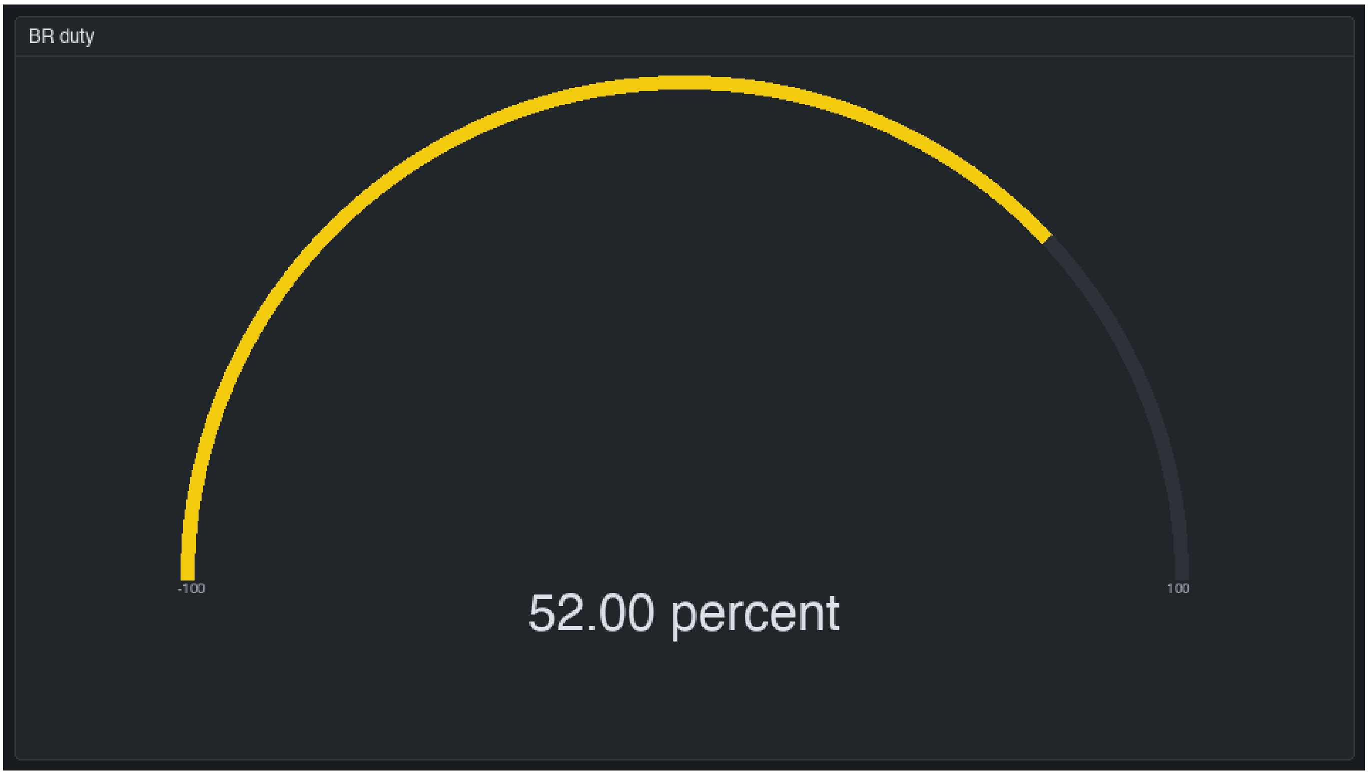
BL duty

Row 0: SLAM map



BR duty

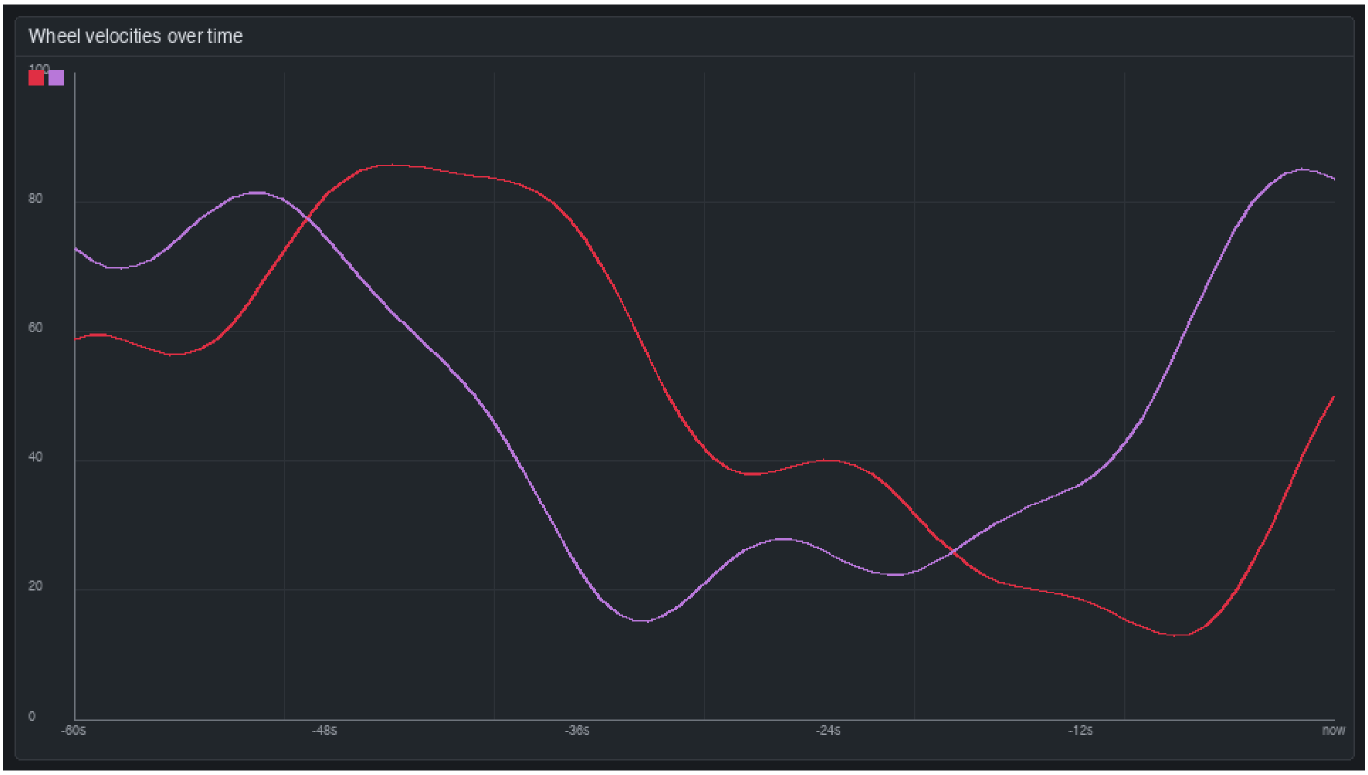
Row 0: SLAM map



Row 1: Motors

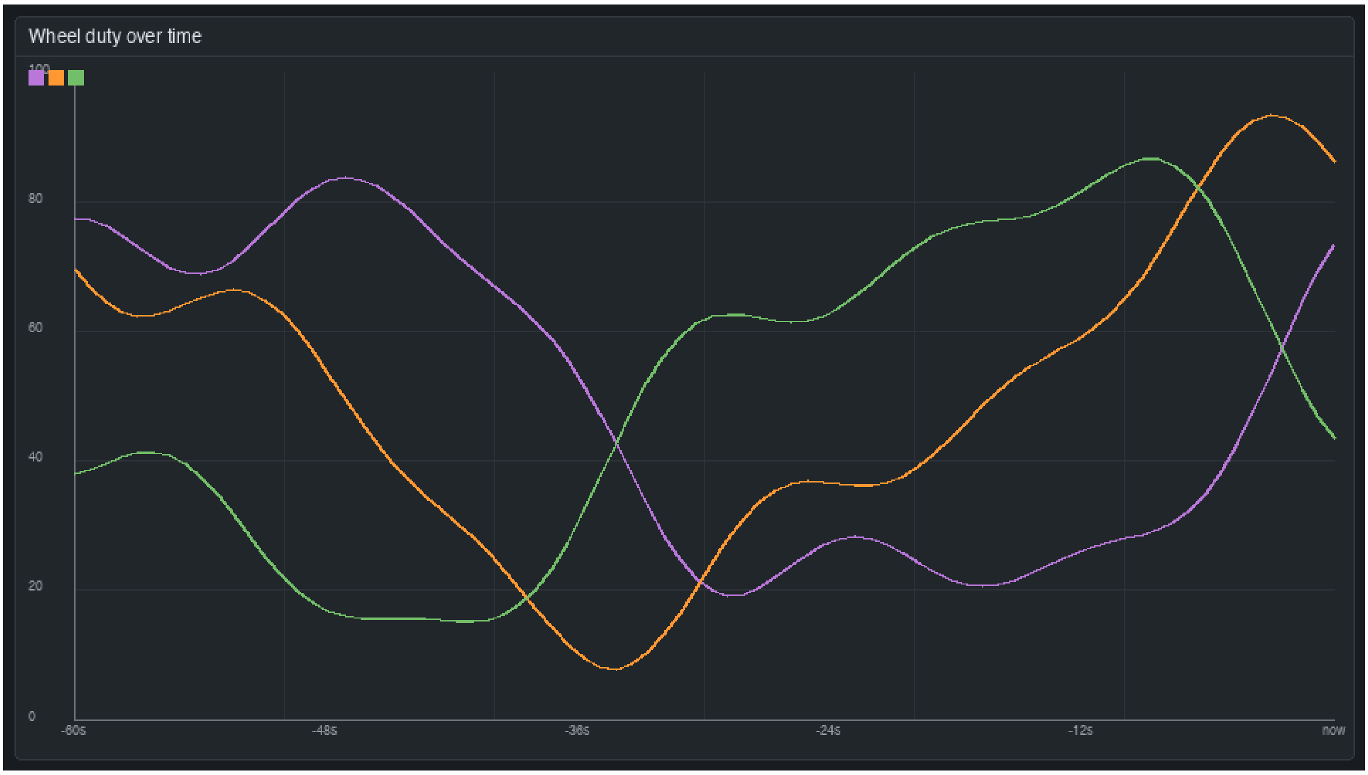
Wheel velocities over time

Row 1: Motors



Wheel duty over time

Row 1: Motors



Motor faults (any != 0 is bad)

Row 1: Motors

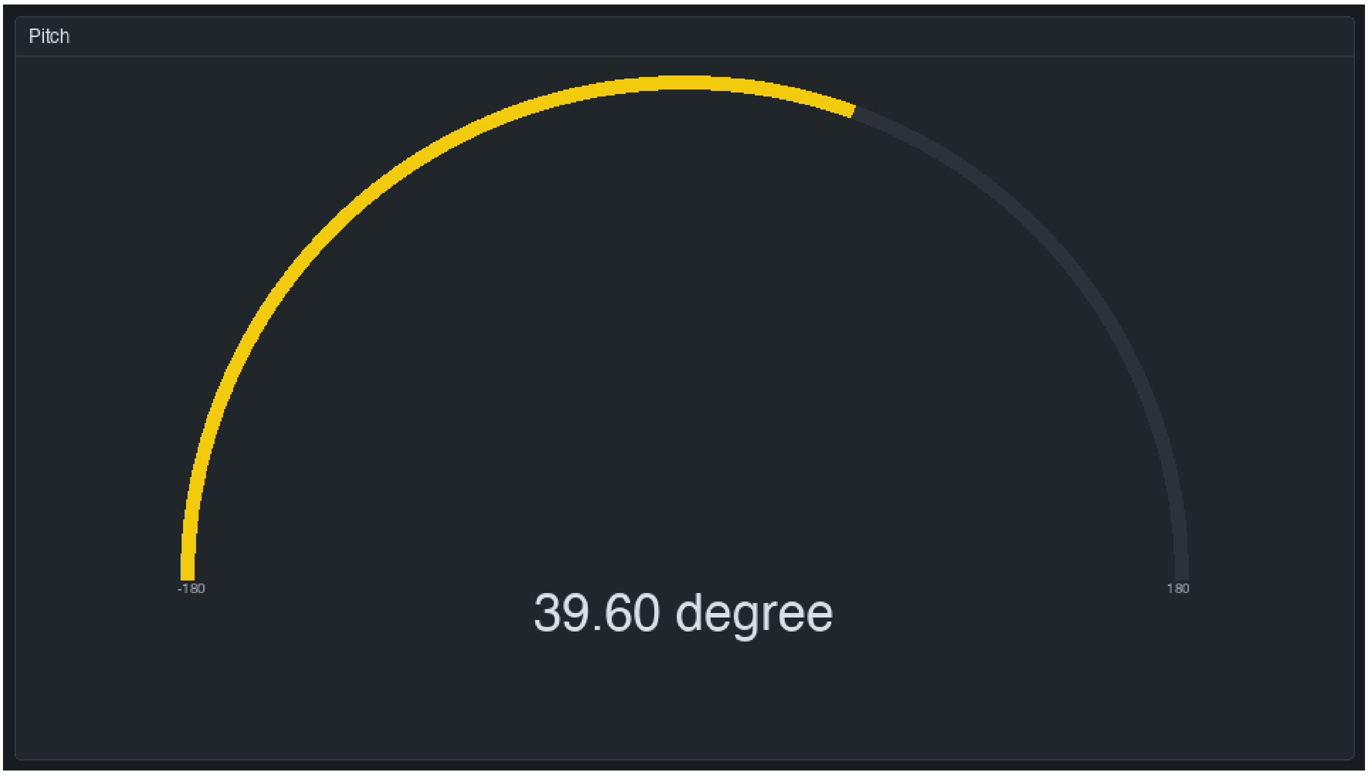
Row 2: IMU

Roll

Row 2: IMU

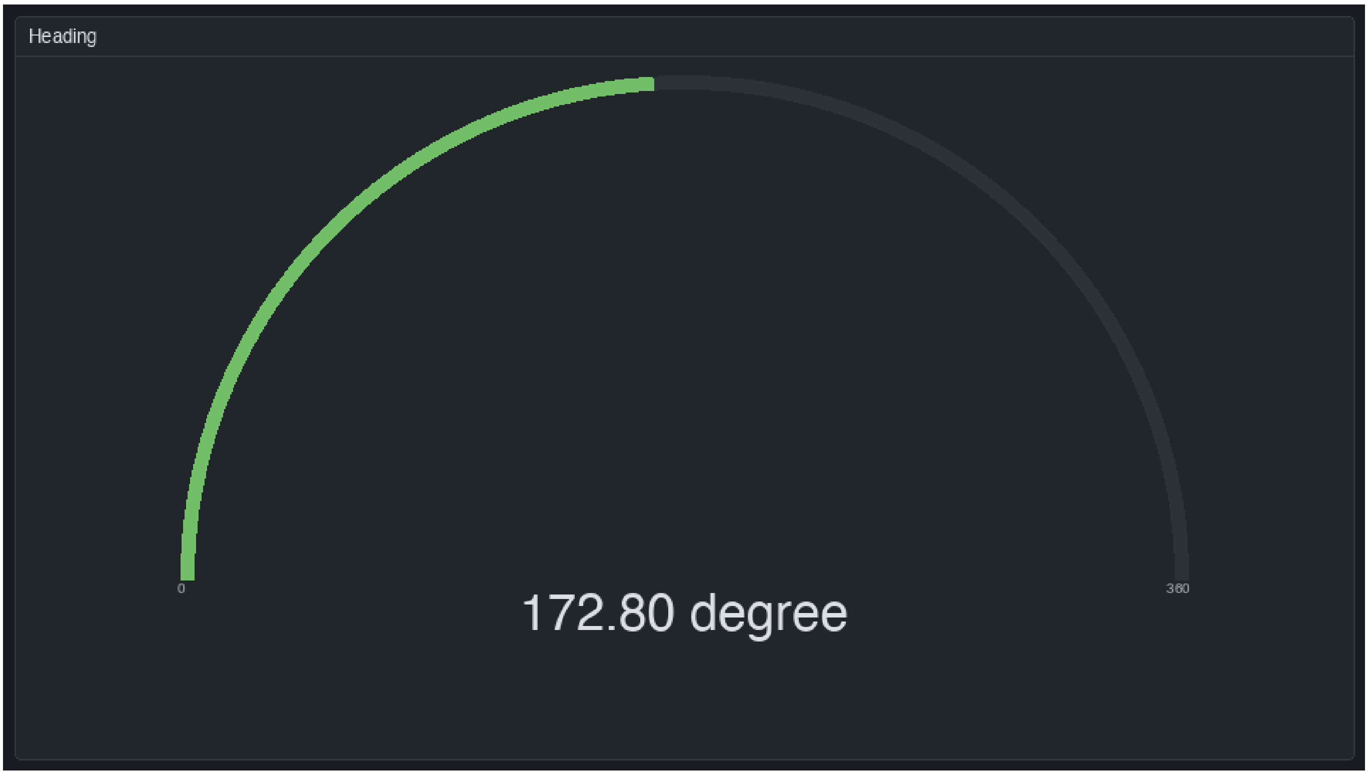
Pitch

Row 2: IMU



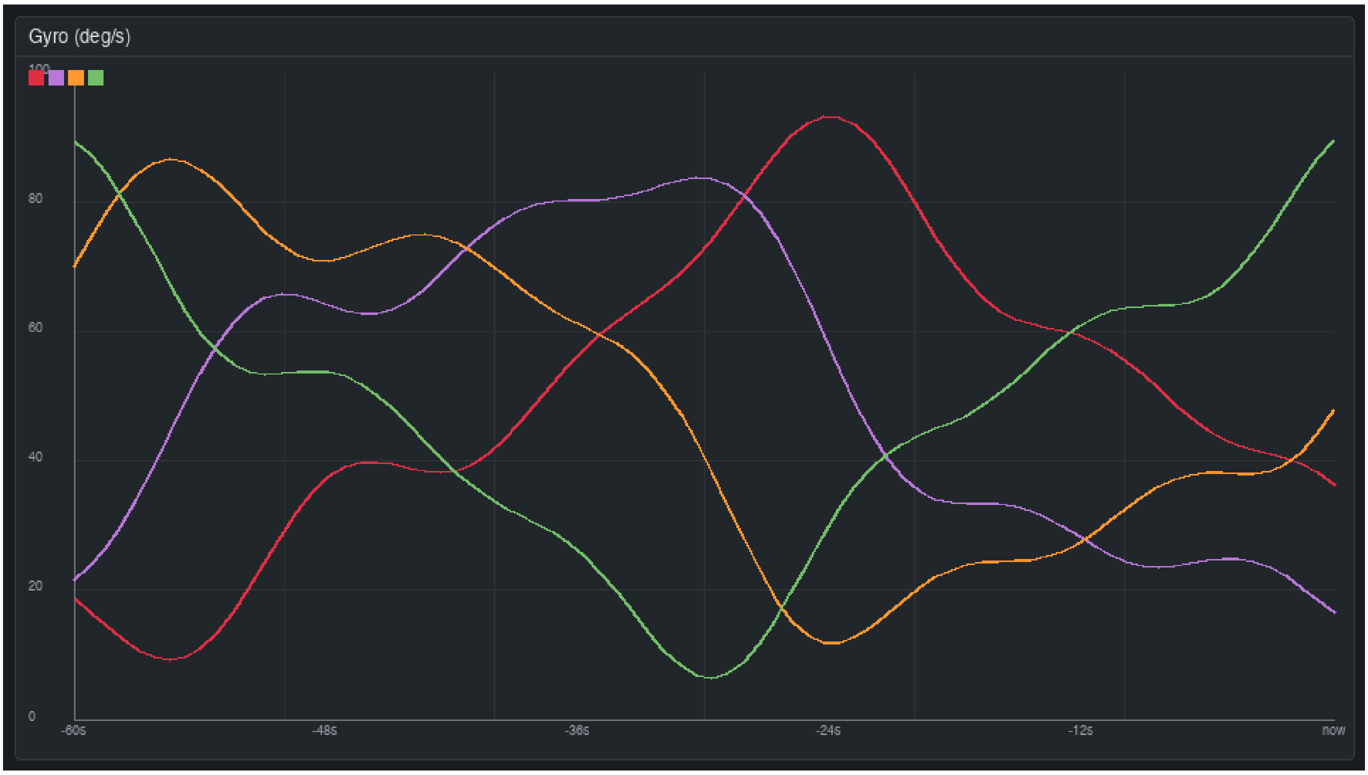
Heading

Row 2: IMU



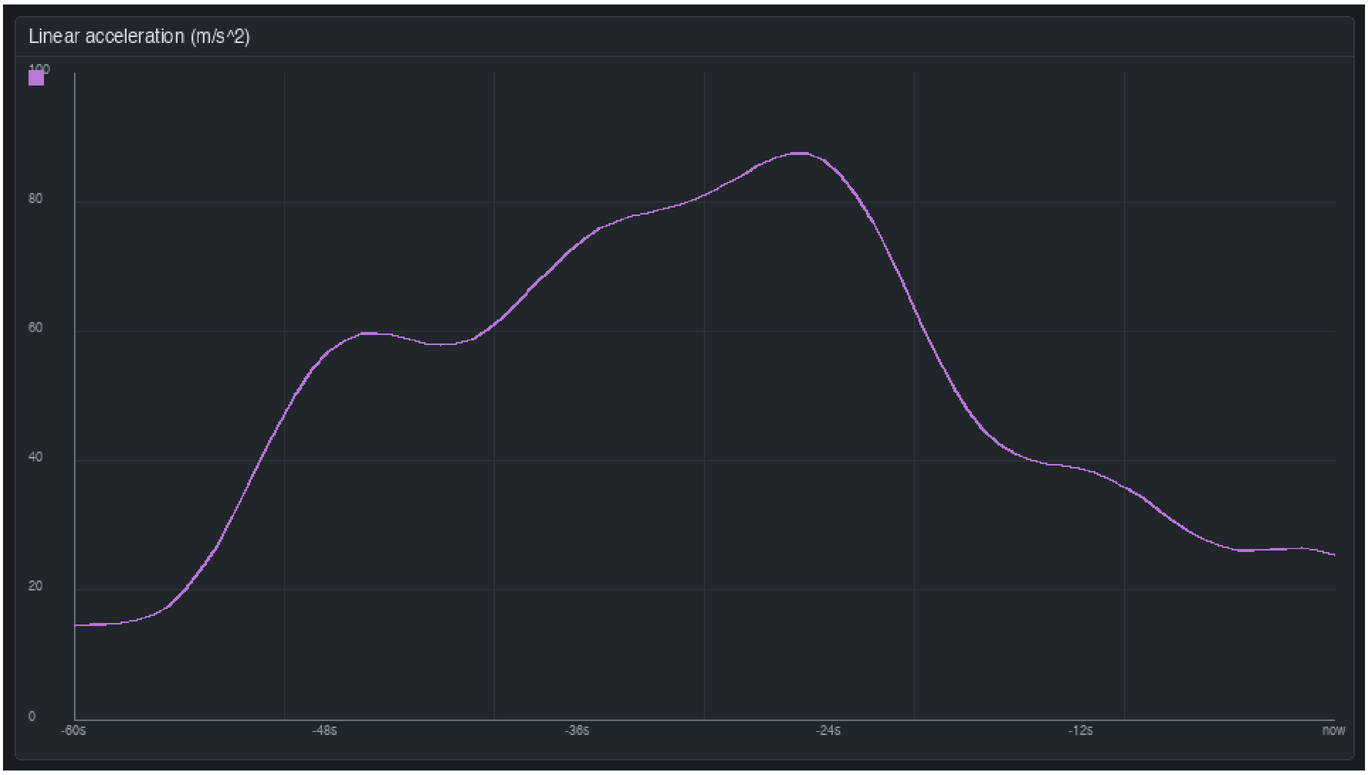
Gyro (deg/s)

Row 2: IMU



Linear acceleration (m/s{}2)

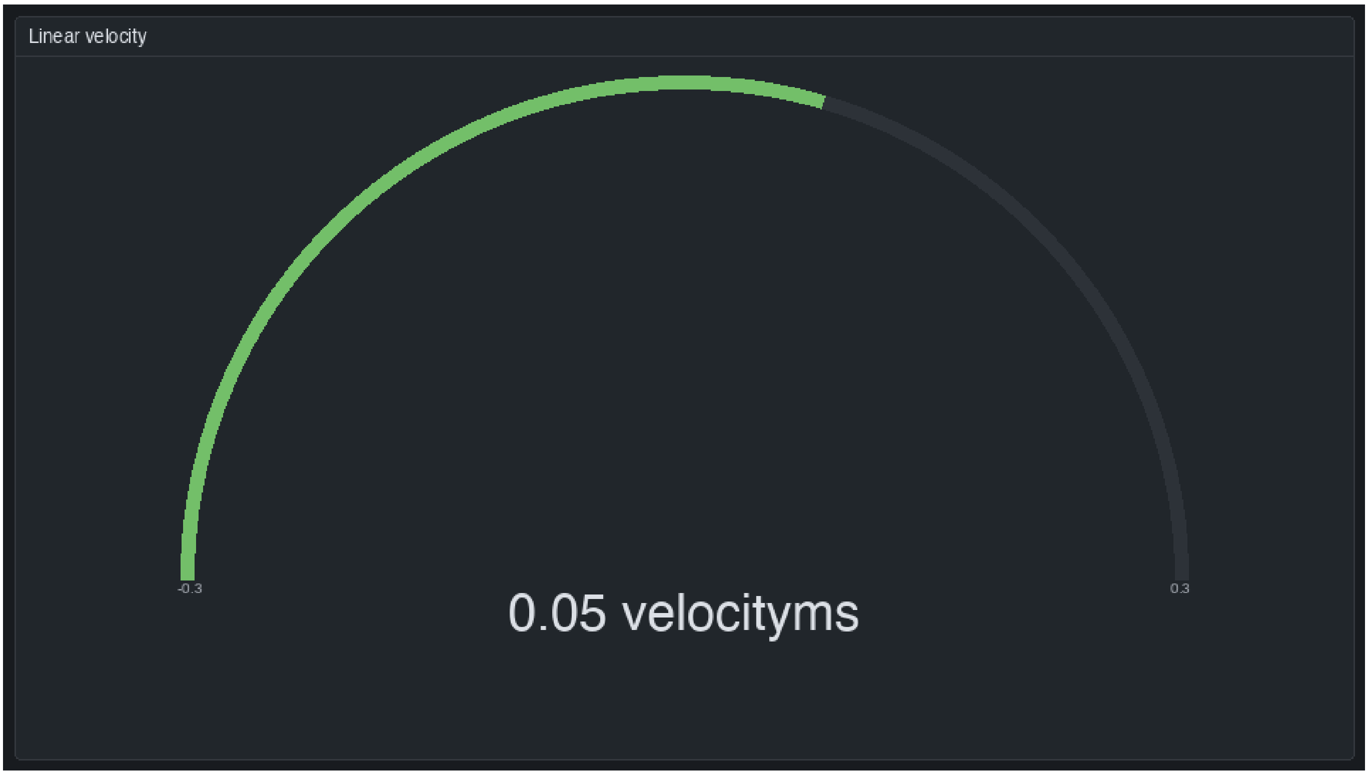
Row 2: IMU



Row 3: Odometry

Linear velocity

Row 3: Odometry

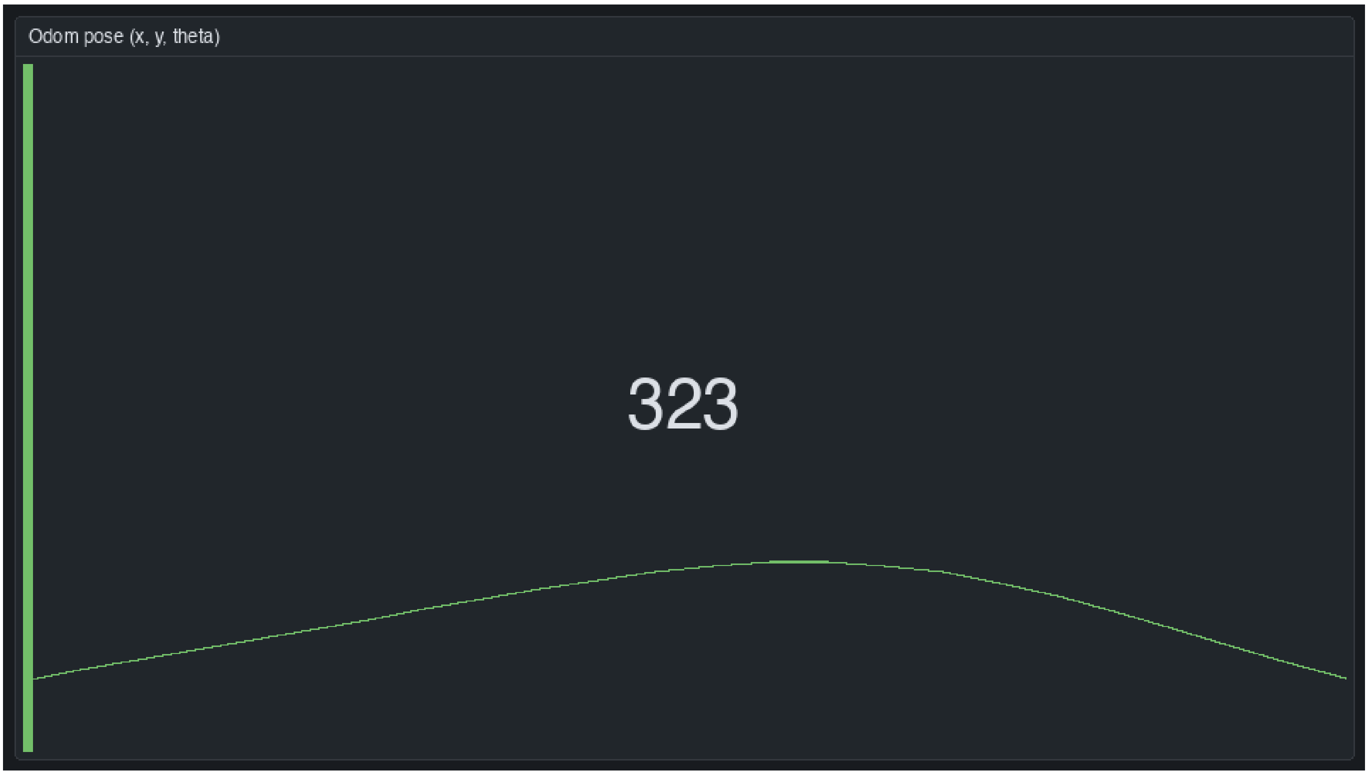


Angular velocity

Row 3: Odometry

Odom pose (x, y, theta)

Row 3: Odometry



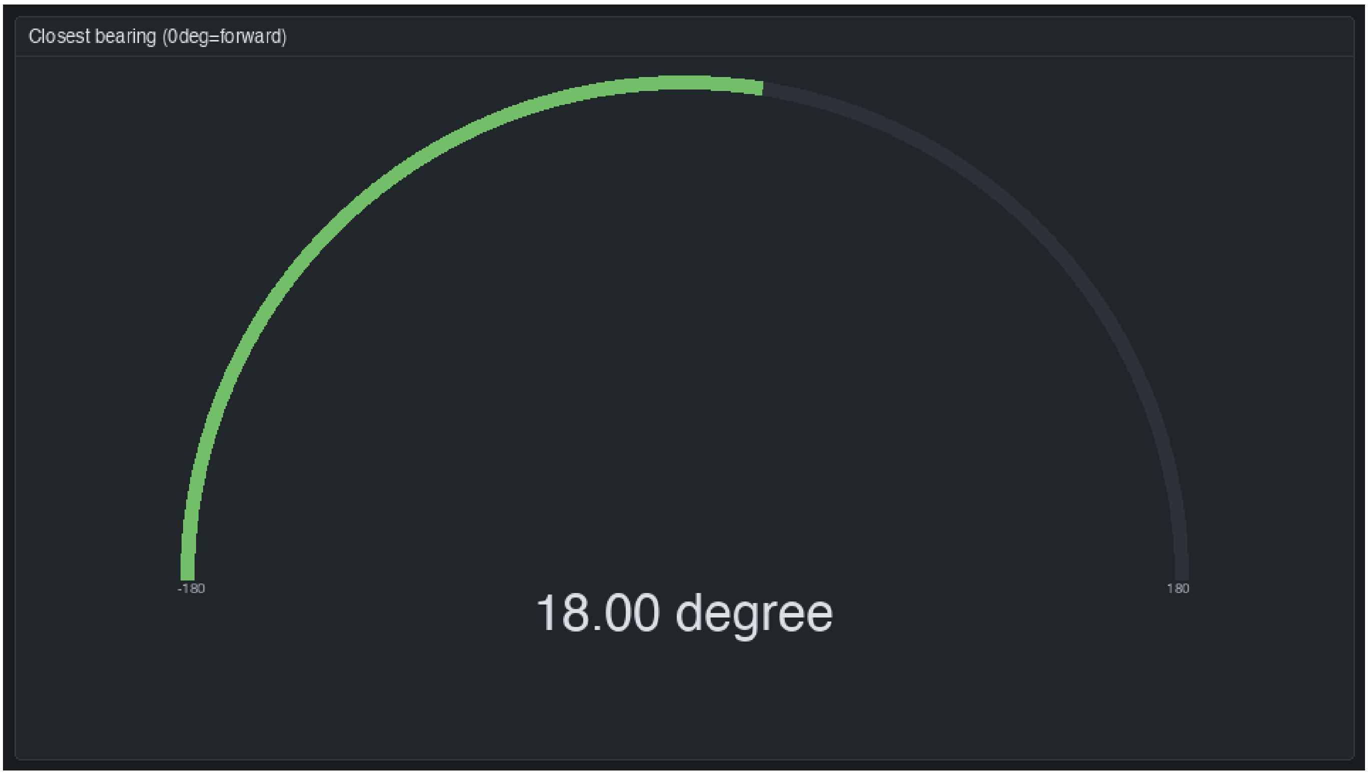
Row 4: LiDAR

Closest obstacle

Row 4: LiDAR

Closest bearing (0deg=forward)

Row 4: LiDAR

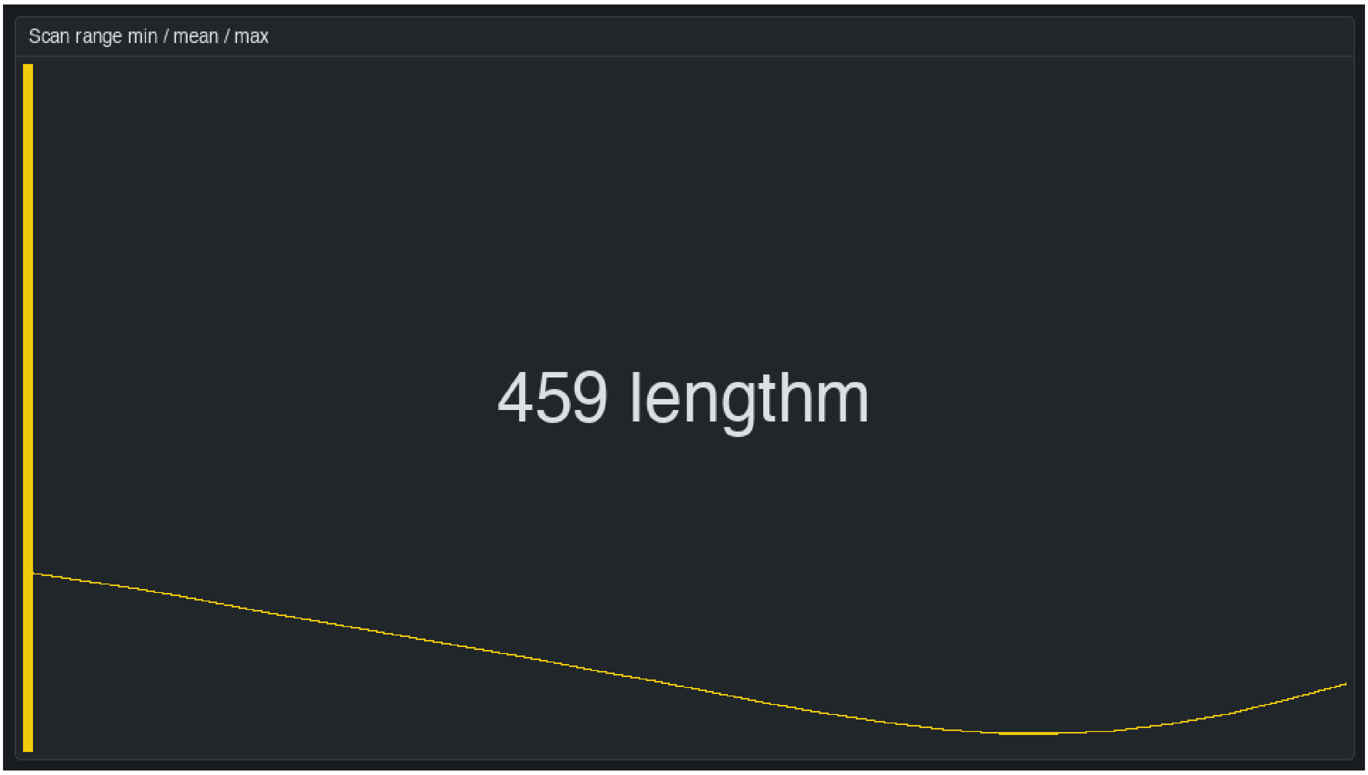


Obstacles < 1 m

Row 4: LiDAR

Scan range min / mean / max

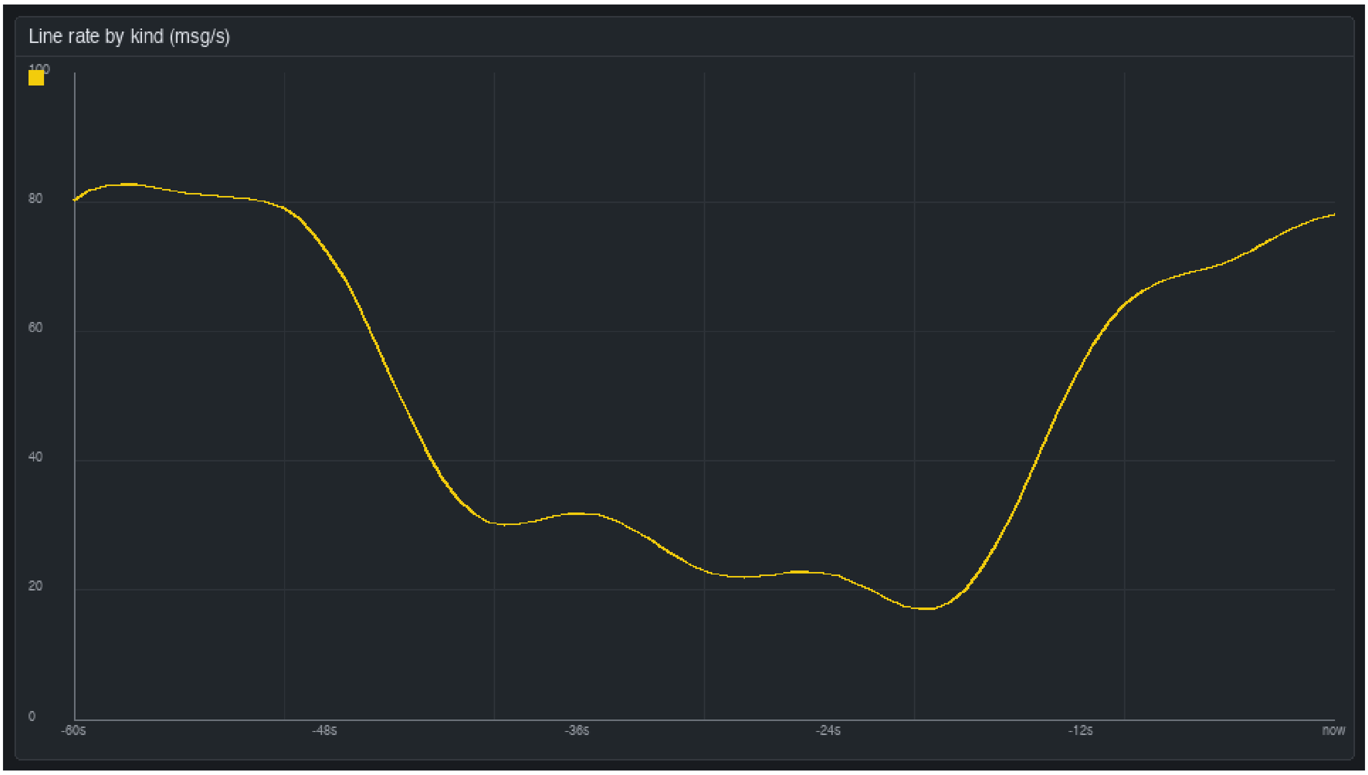
Row 4: LiDAR



Row 5: Bridge health

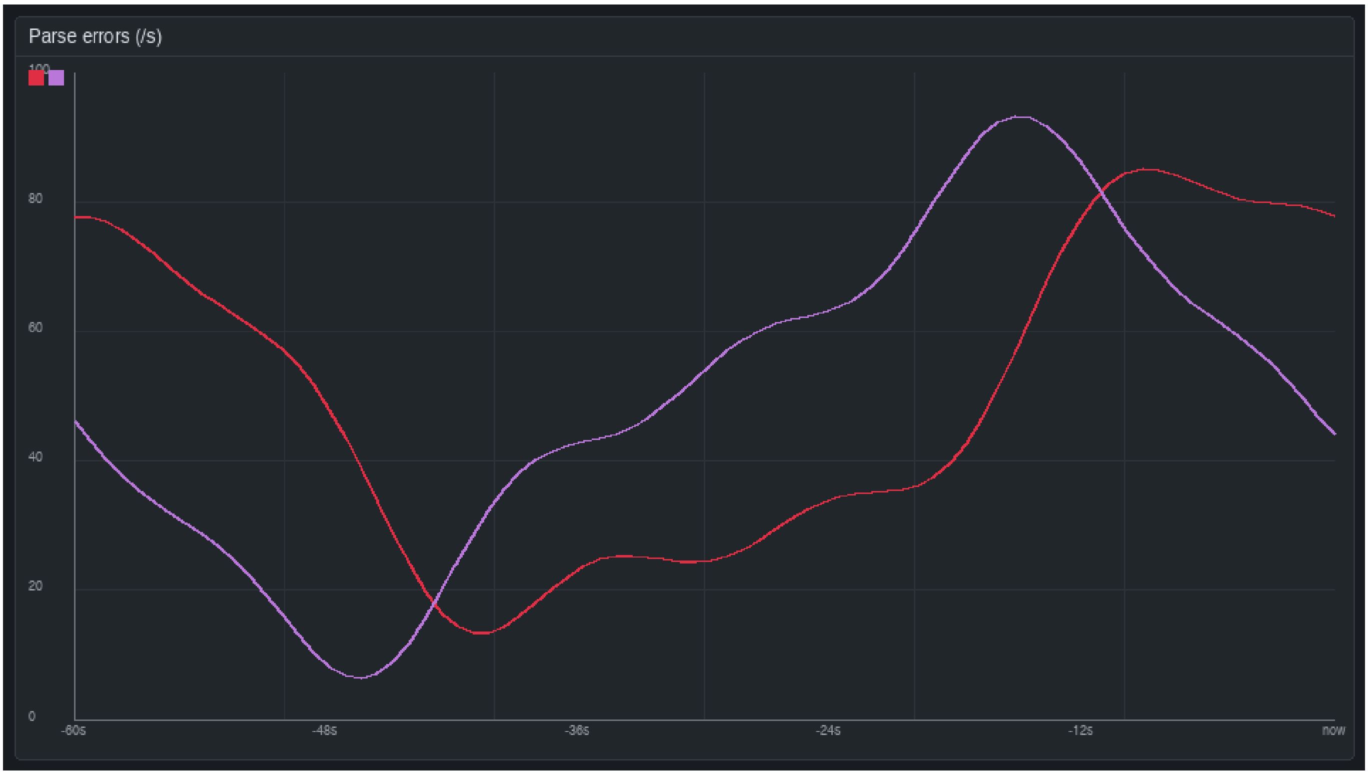
Line rate by kind (msg/s)

Row 5: Bridge health



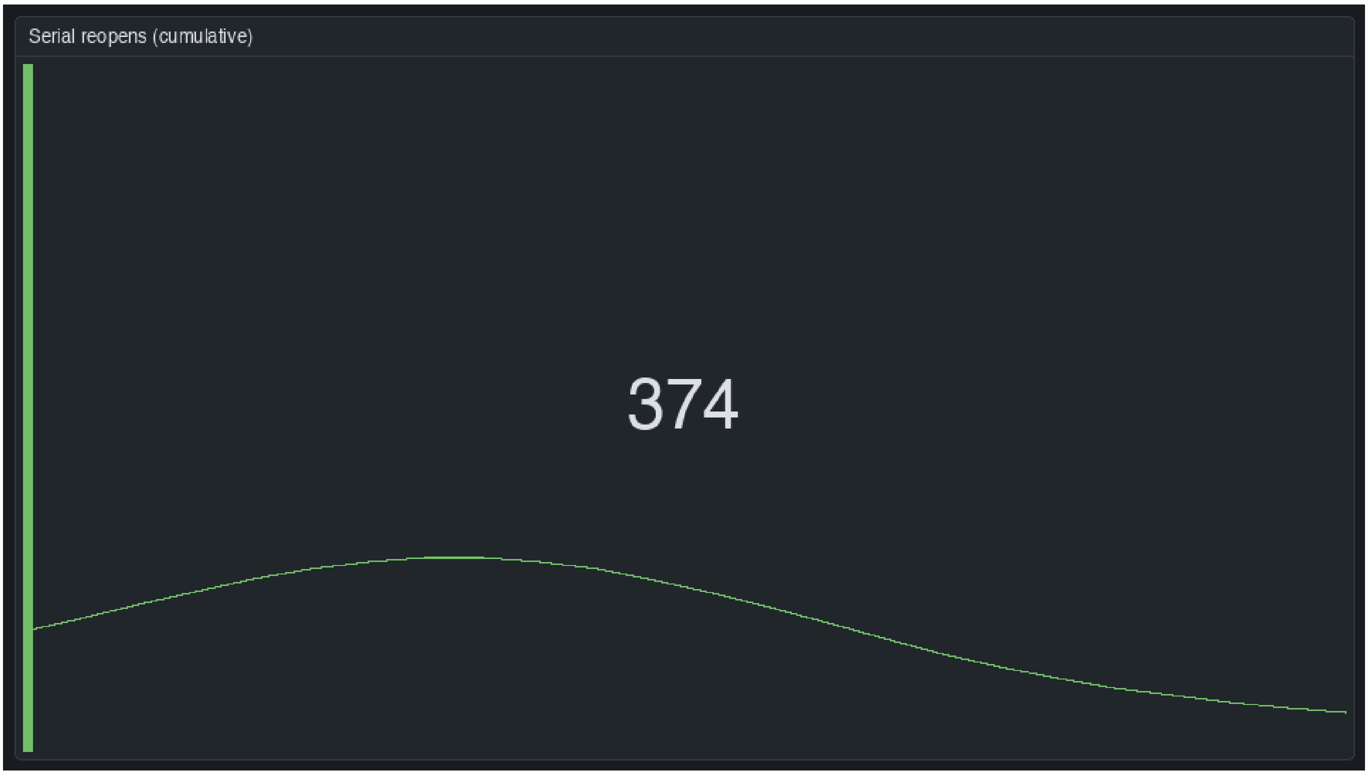
Parse errors (/s)

Row 5: Bridge health



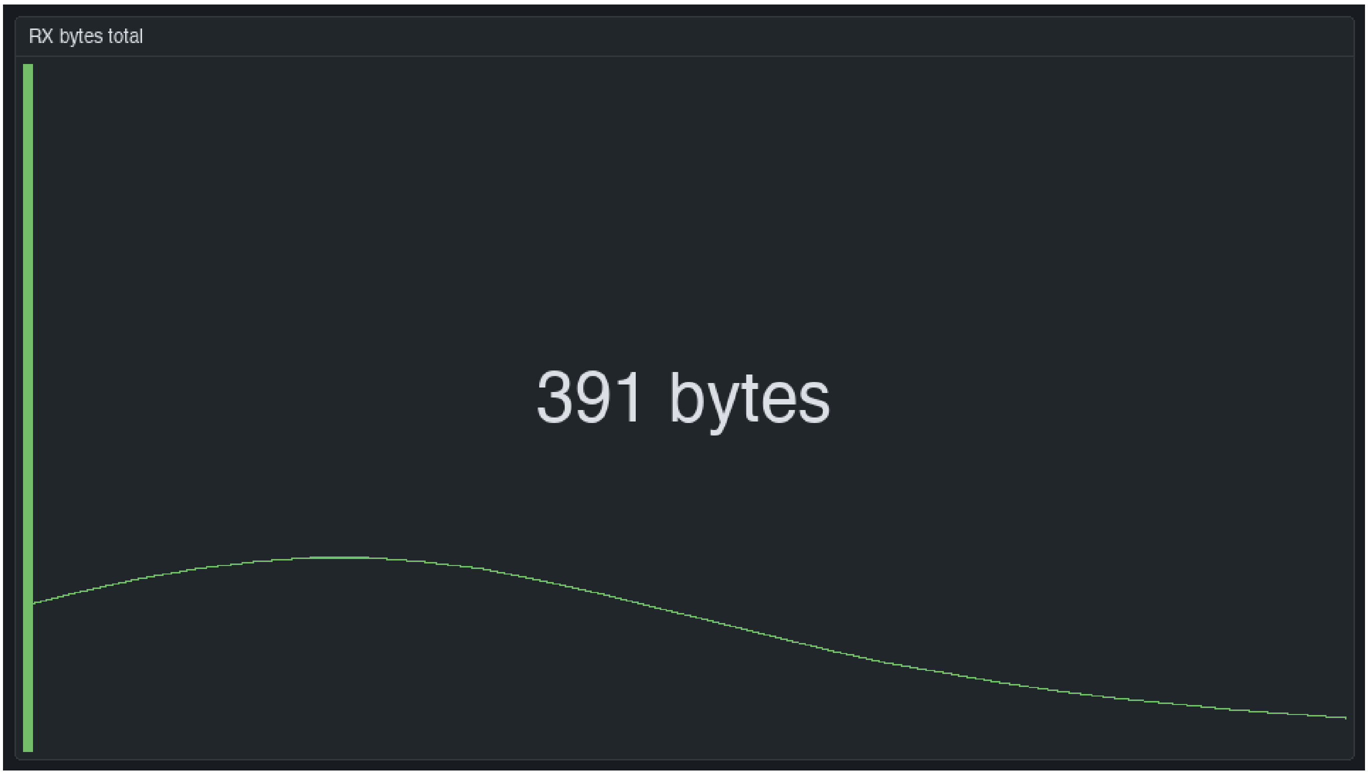
Serial reopens (cumulative)

Row 5: Bridge health



RX bytes total

Row 5: Bridge health



STAR Pi5 Cockpit + Systemd Units

Source Listing

Auto-generated syntax-highlighted listing

Version 1.1
April 2026

STAR Development Team
Locked Inc.

Generated by `scripts/codedump-pdf.sh`.

Contents

1 (root)

README.md

````
# STAR infra -- Pi5-side system files

Non-ROS infra I manage on the Pi5 itself (outside the ROS workspace). These
are mirrored here so they live in version control, but the authoritative
copies are the ones deployed at the paths below.

## Layout

```
infra/pi5/
+-- opt/
|   +-- star_cockpit_api/
|       +-- server.py              -> /opt/star_cockpit_api/server.py
+-- etc/
    +-- caddy/
    |   +-- Caddyfile              -> /etc/caddy/Caddyfile
    +-- systemd/system/
        +-- caddy.service          -> /etc/systemd/system/caddy.service
        +-- star-cockpit-api.service -> /etc/systemd/system/star-cockpit-api.service
        +-- star-slam-mvp.service  -> /etc/systemd/system/star-slam-mvp.service
```

## What each piece does

### `star_cockpit_api` -- HTTP API backing the Grafana cockpit controls

A standalone Python rclpy node + HTTP server on `100.64.0.6:9102`.

Endpoints:

| Method | Path                        | Purpose                                                |
|--------|-----------------------------|--------------------------------------------------------|
| GET    | `/api/state`                | `{autonomy, estop, speed, slam}` -- all cached         |
| POST   | `/api/autonomy/toggle`      | flip `/star/autonomy_enable`                           |
| POST   | `/api/estop/toggle`         | flip `/star/estop`                                     |
| POST   | `/api/speed`                | body `{"value":0.75}` -> publish `/star/speed_scale`   |
| POST   | `/api/cmd_vel`              | body `{"linear":0.5,"angular":0}` -> publish `/cmd_vel` |
| POST   | `/api/slam/start`           | configure+activate `/slam_toolbox`                     |
| POST   | `/api/slam/stop`            | deactivate `/slam_toolbox`                             |

Subscribes to the latched `/star/state/*` topics so the toggle logic always
sees the supervisor's current view. A background thread refreshes the slam
lifecycle state every 4s (via `ros2 service call`) so `/api/state` is O(1).

### Systemd units

- **`caddy.service`** -- Caddy (on Pi5) listens on `100.64.0.6:443`, terminates
  TLS with its own local CA (`tls internal`), fronts `star_simple_bridge` on
  `:9101` and `star_cockpit_api` on `:9102`. `After=tailscaled.service` so it
  waits for the tailscale IP to be assigned. `Restart=always`.

- **`star-cockpit-api.service`** -- runs `server.py` as the `star` user, with
  ROS workspace sourced and Cyclone DDS config exported. Restart on failure.

- **`star-slam-mvp.service`** -- runs `scripts/slam-mvp.sh start` at boot
  (after tailscale is up + an 8s USB-enumeration delay). Initial state is
  MANUAL mode, e-stop cleared -- so the robot is safe on boot, operator
  flips to AUTONOMY via the Grafana cockpit toggle.

### `Caddyfile`

Pi5-side Caddy config. Uses `tls internal` -- no ACME, no Cloudflare, no
external CA. Laptop/phone clients install Caddy's local root CA once (see
`docs/ARCHITECTURE.md` -> "Client setup to use `star.local`"). No secrets.

## Installing on a fresh Pi5

```bash
# 1. Copy files (from repo root)
sudo cp infra/pi5/opt/star_cockpit_api/server.py /opt/star_cockpit_api/
sudo chmod +x /opt/star_cockpit_api/server.py

sudo cp infra/pi5/etc/caddy/Caddyfile /etc/caddy/Caddyfile
sudo cp infra/pi5/etc/systemd/system/*.service /etc/systemd/system/

````
````

````
# 2. Install Caddy binary with no plugins needed (tls internal only)
sudo curl -fsSL -o /usr/local/bin/caddy \
  "https://caddyserver.com/api/download?os=linux&arch=arm64"
sudo chmod +x /usr/local/bin/caddy
sudo useradd --system --shell /usr/sbin/nologin --home-dir /var/lib/caddy caddy 2>/dev/null || true
sudo mkdir -p /var/log/caddy /var/lib/caddy
sudo chown caddy:caddy /var/log/caddy /var/lib/caddy

# 3. Enable + start
sudo systemctl daemon-reload
sudo systemctl enable --now caddy star-cockpit-api star-slam-mvp
```

## What is deliberately NOT mirrored here

- **Cluster-side k8s manifests** (traefik, robot-gateway, lichtblick, grafana
  dashboard JSON). Those live in k3s's etcd; Grafana dashboard is pushed via
  API. They can be exported separately if needed.
- **Any hostname or credentials** -- these system files use only `star.local`
  and `100.64.0.6` (tailscale IP), nothing that identifies the operator.
- **The Caddy-gateway Caddyfile block on pve** that defines
  `<PUBLIC_ROBOT_HOST>` -- that contains the public hostname and a bcrypt
  password hash, and is managed on the homelab host, not on the Pi5.
````

2 pi5

pi5/etc/systemd/system/caddy.service

```
[Unit]
Description=Caddy
After=network-online.target tailscaled.service
Wants=network-online.target tailscaled.service

[Service]
User=caddy
Group=caddy
ExecStart=/usr/local/bin/caddy run --config /etc/caddy/Caddyfile
ExecReload=/usr/local/bin/caddy reload --config /etc/caddy/Caddyfile --force
Restart=always
RestartSec=5s
TimeoutStopSec=5s
LimitNOFILE=1048576
AmbientCapabilities=CAP_NET_ADMIN CAP_NET_BIND_SERVICE

[Install]
WantedBy=multi-user.target
```

pi5/etc/systemd/system/star-cockpit-api.service

```
[Unit]
Description=STAR cockpit HTTP API (Grafana toggle/speed backend)
After=network-online.target

[Service]
User=star
Group=star
ExecStart=/bin/bash -c "source /opt/ros/jazzy/setup.bash && source /workspaces/STAR/star-ros2/install/setup.bash && export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp && export CYCLONEDDS_URI=file:///workspaces/STAR/config/cyclonedds.xml && exec python3 /opt/star_cockpit_api/server.py"
Restart=always
RestartSec=3

[Install]
WantedBy=multi-user.target
```

pi5/etc/systemd/system/star-slam-mvp.service

```
[Unit]
Description=STAR SLAM MVP stack (supervisor + simple_bridge + LiDAR + SLAM + foxglove)
Documentation=file:///workspaces/STAR/docs/ARCHITECTURE.md
After=network-online.target tailscaled.service dev-star\x2dmcu.device
Wants=network-online.target tailscaled.service

```
```

```
[Service]
Type=forking
User=star
Group=star
Environment=HOME=/home/star
Environment=USER=star
Environment=PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
# Give USB devices time to enumerate after boot
ExecStartPre=/bin/sleep 8
ExecStart=/bin/bash /workspaces/STAR/scripts/slam-mvp.sh start
ExecStop=/bin/bash /workspaces/STAR/scripts/slam-mvp.sh stop
# The script backgrounds each node and returns; forking type matches
RemainAfterExit=yes
Restart=no

[Install]
WantedBy=multi-user.target
```

pi5/opt/star_cockpit_api/server.py

```python
#!/usr/bin/env python3
"""HTTP API backing the Grafana cockpit controls.

  GET  /api/state             -> {"autonomy": bool|null, "estop": bool|null, "speed": float|null}
  POST /api/autonomy/toggle   -> flips /star/autonomy_enable, returns new state
  POST /api/estop/toggle      -> flips /star/estop, returns new state
  POST /api/speed  body: {"value": 0.75}  -> publishes /star/speed_scale, returns applied

Binds to 127.0.0.1:9102; Caddy fronts it as https://star.local/api/*
State is learned from the supervisor's latched /star/state/* topics.
"""
import json
import subprocess
import re
import threading
from http.server import BaseHTTPRequestHandler, ThreadingHTTPServer
import rclpy
from rclpy.node import Node
from rclpy.qos import QoSProfile, DurabilityPolicy, ReliabilityPolicy, HistoryPolicy
from std_msgs.msg import Bool, Float32
from geometry_msgs.msg import Twist

PORT = 9102
LATCHED = QoSProfile(
    history=HistoryPolicy.KEEP_LAST, depth=1,
    reliability=ReliabilityPolicy.RELIABLE,
    durability=DurabilityPolicy.TRANSIENT_LOCAL,
)


class CockpitNode(Node):
    def __init__(self):
        super().__init__("star_cockpit_api")
        self._lock = threading.Lock()
        self._autonomy = None
        self._estop = None
        self._last_speed = None
        self._slam_state_cached = None
        self._autonomy_pub = self.create_publisher(Bool, "/star/autonomy_enable", 10)
        self._estop_pub = self.create_publisher(Bool, "/star/estop", 10)
        self._speed_pub = self.create_publisher(Float32, "/star/speed_scale", 10)
        self._cmd_vel_pub = self.create_publisher(Twist, "/cmd_vel", 10)
        self.create_subscription(Bool, "/star/state/autonomy_enable", self._on_auto, LATCHED)
        self.create_subscription(Bool, "/star/state/estop", self._on_estop, LATCHED)
        self.get_logger().info(f"cockpit HTTP on 127.0.0.1:{PORT}")

    def _on_auto(self, m):
        with self._lock: self._autonomy = bool(m.data)

    def _on_estop(self, m):
        with self._lock: self._estop = bool(m.data)

    def get_state(self):
        with self._lock:
            return {"autonomy": self._autonomy, "estop": self._estop, "speed": self._last_speed, "slam":
            ↪  self.slam_state()}

    def toggle_autonomy(self):
```

```
        with self._lock:
            new = not bool(self._autonomy)
        self._autonomy_pub.publish(Bool(data=new))
        return new

    def toggle_estop(self):
        with self._lock:
            new = not bool(self._estop)
        self._estop_pub.publish(Bool(data=new))
        return new

    def slam_lifecycle(self, transition_id):
        try:
            r = subprocess.run(
                ["ros2","service","call","/slam_toolbox/change_state",
                 "lifecycle_msgs/srv/ChangeState",
                 "{transition: {id: %d}}" % transition_id],
                capture_output=True, text=True, timeout=15)
            ok = "success=True" in r.stdout
            self._refresh_slam_state()
            return ok
        except Exception:
            return False

    def slam_state(self):
        return self._slam_state_cached

    def _refresh_slam_state(self):
        try:
            r = subprocess.run(
                ["ros2","service","call","/slam_toolbox/get_state",
                 "lifecycle_msgs/srv/GetState"],
                capture_output=True, text=True, timeout=5)
            m = re.search(r"label=\'([a-z_]+)\'", r.stdout)
            new = m.group(1) if m else None
            with self._lock:
                self._slam_state_cached = new
        except Exception:
            pass

    def set_cmd_vel(self, linear, angular):
        m = Twist(); m.linear.x = float(linear); m.angular.z = float(angular)
        self._cmd_vel_pub.publish(m)
        return {"linear": m.linear.x, "angular": m.angular.z}

    def set_speed(self, v):
        v = max(0.0, min(1.0, float(v)))
        with self._lock:
            self._last_speed = v
        self._speed_pub.publish(Float32(data=v))
        return v


NODE = None  # set in main


class H(BaseHTTPRequestHandler):
    def log_message(self, *a, **k): pass

    def _json(self, code, obj):
        body = json.dumps(obj).encode()
        self.send_response(code)
        self.send_header("Content-Type", "application/json")
        self.send_header("Content-Length", str(len(body)))
        self.send_header("Cache-Control", "no-store")
        self.send_header("Access-Control-Allow-Origin", "*")
        self.send_header("Access-Control-Allow-Methods", "GET, POST, OPTIONS")
        self.send_header("Access-Control-Allow-Headers", "Content-Type")
        self.end_headers()
        self.wfile.write(body)

    def do_OPTIONS(self):
        self._json(204, {})

    def do_GET(self):
        if self.path == "/api/state":
            return self._json(200, NODE.get_state())
        self._json(404, {"error": "not found"})
```

```
    def do_POST(self):
        if self.path == "/api/autonomy/toggle":
            return self._json(200, {"autonomy": NODE.toggle_autonomy()})
        if self.path == "/api/estop/toggle":
            return self._json(200, {"estop": NODE.toggle_estop()})
        if self.path == "/api/slam/start":
            ok = NODE.slam_lifecycle(3)
            if not ok:
                NODE.slam_lifecycle(1)
                ok = NODE.slam_lifecycle(3)
            return self._json(200 if ok else 500, {"slam": NODE.slam_state(), "ok": ok})
        if self.path == "/api/slam/stop":
            ok = NODE.slam_lifecycle(4)
            return self._json(200 if ok else 500, {"slam": NODE.slam_state(), "ok": ok})
        if self.path == "/api/cmd_vel":
            n = int(self.headers.get("Content-Length", 0) or 0)
            try:
                body = json.loads(self.rfile.read(n).decode() or "{}")
                lin = float(body.get("linear", 0))
                ang = float(body.get("angular", 0))
            except Exception as e:
                return self._json(400, {"error": f"bad body: {e}"})
            return self._json(200, NODE.set_cmd_vel(lin, ang))
        if self.path == "/api/speed":
            n = int(self.headers.get("Content-Length", 0) or 0)
            try:
                v = float(json.loads(self.rfile.read(n).decode() or "{}").get("value"))
            except Exception as e:
                return self._json(400, {"error": f"bad body: {e}"})
            return self._json(200, {"speed": NODE.set_speed(v)})
        self._json(404, {"error": "not found"})


def _slam_refresher_loop():
    import time
    while True:
        try:
            if NODE is not None:
                NODE._refresh_slam_state()
        except Exception:
            pass
        time.sleep(4)

def main():
    global NODE
    rclpy.init()
    NODE = CockpitNode()
    srv = ThreadingHTTPServer(("100.64.0.6", PORT), H)
    threading.Thread(target=srv.serve_forever, name="cockpit-http", daemon=True).start()
    threading.Thread(target=_slam_refresher_loop, name="slam-refresh", daemon=True).start()
    try:
        rclpy.spin(NODE)
    finally:
        NODE.destroy_node()
        rclpy.shutdown()


if __name__ == "__main__":
    main()
```